Import Data using Transform Maps (Spreadsheet)

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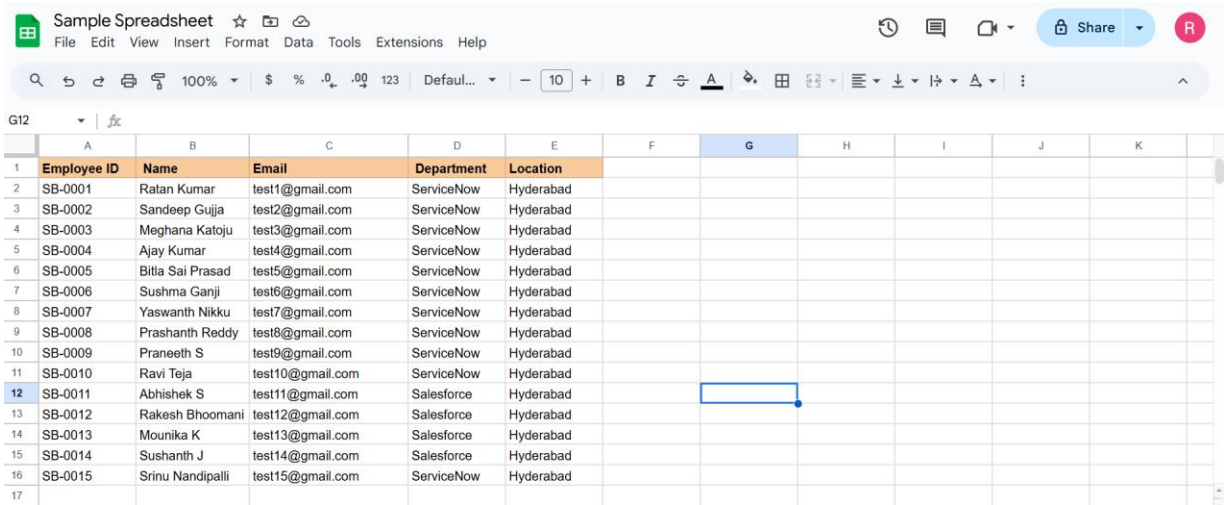
Description:

This micro project focuses on importing structured data from an external spreadsheet into the **ServiceNow**platformusingImportSetsandTransformMaps.Theprojectsimulatesareal-world scenario where bulk employee data is received in Excel format and must be migrated into ServiceNowefficientlyandaccurately.ThespreadsheetdataisfirstloadedintoanImportSet table, which serves as a staging area, and a Transform Map is then configured to map source fields to the appropriate target table fields. Upon executing the transformation, records are created in the target table and validated to ensure data integrity. This project helps beginners understand the end-to-end data import process in ServiceNow, including data loading, field mapping, transformation execution, and result verification.

Step by Step process:

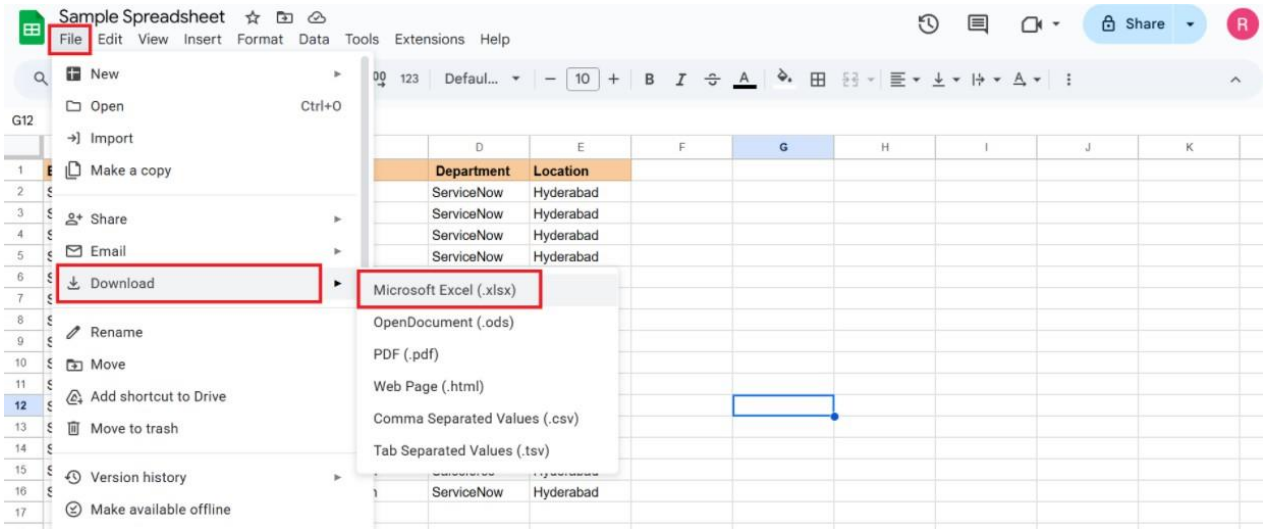
Phase1: Prepare the sheets

1. Open New Google Spreadsheet.
2. Create New Google Spreadsheet with sample data as shown below.



	A	B	C	D	E	F	G	H	I	J	K
1	**Employee ID**	**Name**	**Email**	**Department**	**Location**						
2	SB-0001	Ratan Kumar	test1@gmail.com	ServiceNow	Hyderabad						
3	SB-0002	Sandeep Gujja	test2@gmail.com	ServiceNow	Hyderabad						
4	SB-0003	Meghana Katoju	test3@gmail.com	ServiceNow	Hyderabad						
5	SB-0004	Ajay Kumar	test4@gmail.com	ServiceNow	Hyderabad						
6	SB-0005	Bitla Sai Prasad	test5@gmail.com	ServiceNow	Hyderabad						
7	SB-0006	Sushma Ganji	test6@gmail.com	ServiceNow	Hyderabad						
8	SB-0007	Yaswanth Nikku	test7@gmail.com	ServiceNow	Hyderabad						
9	SB-0008	Prashanth Reddy	test8@gmail.com	ServiceNow	Hyderabad						
10	SB-0009	Praneeth S	test9@gmail.com	ServiceNow	Hyderabad						
11	SB-0010	Ravi Teja	test10@gmail.com	ServiceNow	Hyderabad						
12	SB-0011	Abhishek S	test11@gmail.com	Salesforce	Hyderabad						
13	SB-0012	Rakesh Bhoomani	test12@gmail.com	Salesforce	Hyderabad						
14	SB-0013	Mounika K	test13@gmail.com	Salesforce	Hyderabad						
15	SB-0014	Sushanth J	test14@gmail.com	Salesforce	Hyderabad						
16	SB-0015	Srinu Nandipalli	test15@gmail.com	ServiceNow	Hyderabad						
17											

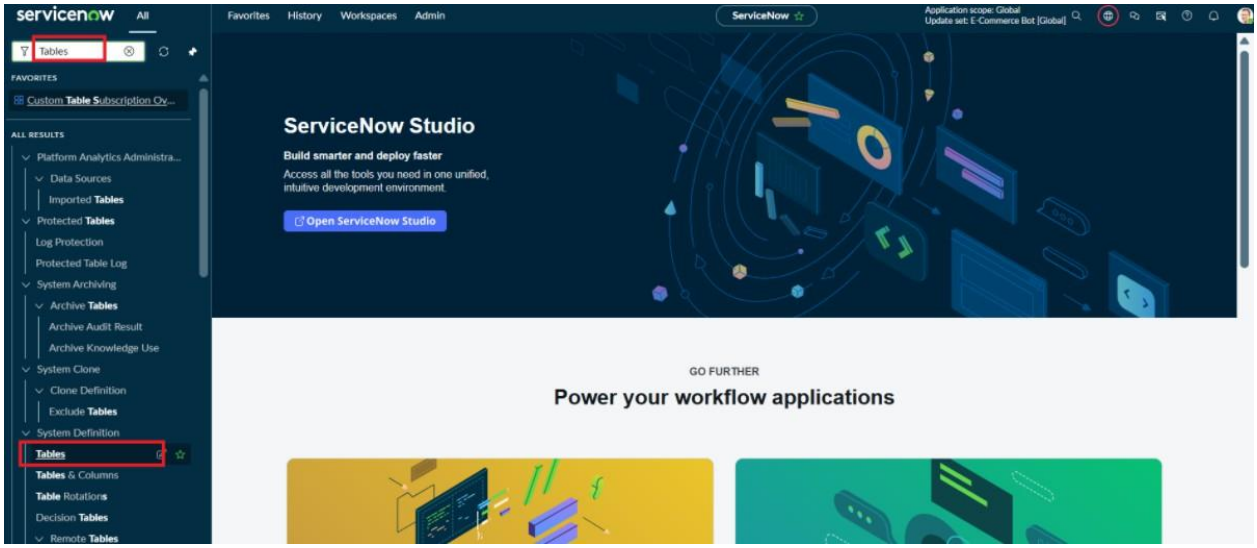
3. Download the spread sheet as Sample Spreadsheet.xlsx format.



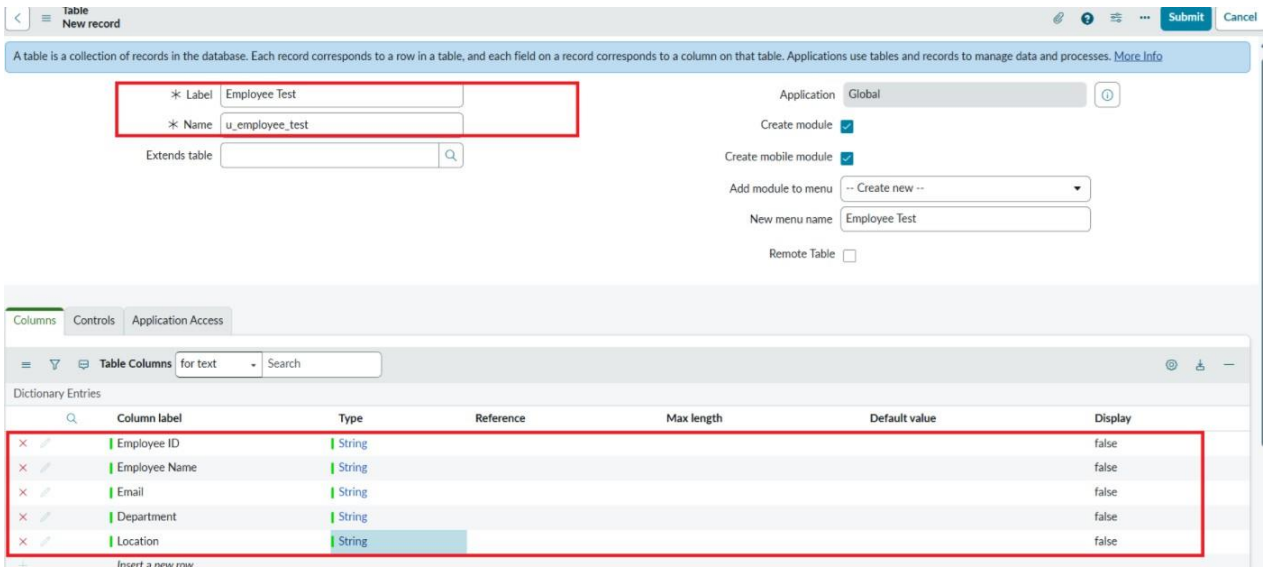
Phase - 2: Create new Custom table

Navigation:

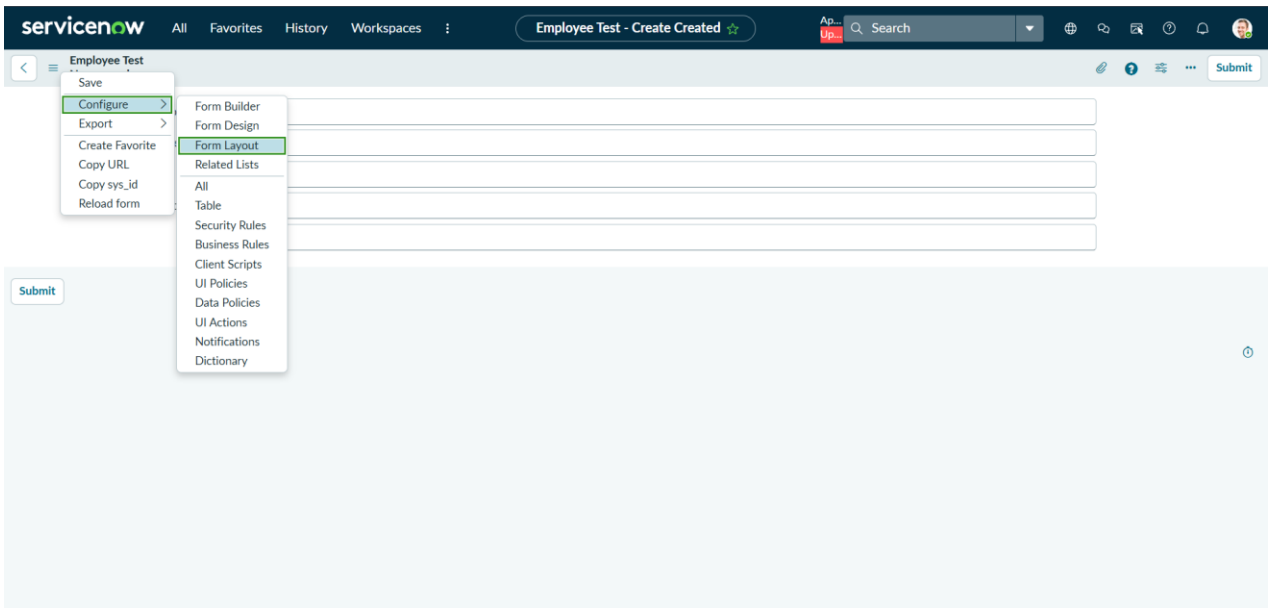
- Tables > Create New



- Label: Employee Test
- Name: u_employee_test
- After creating and saving the **Employee Test** table, **scroll down** to the bottom of the table form.

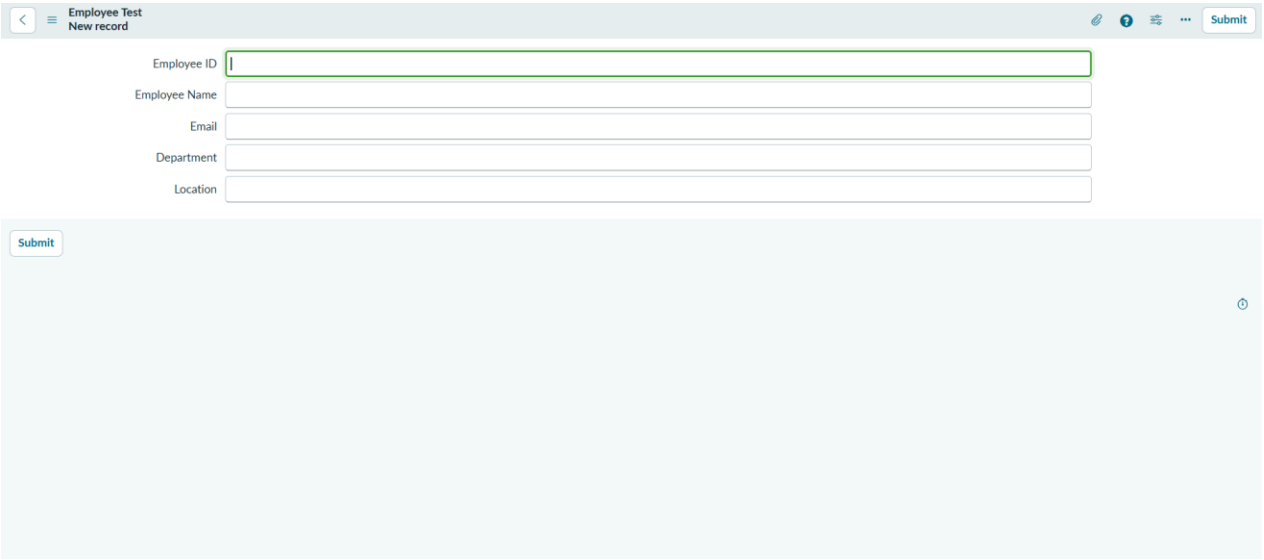


- Click **Show Form** to open the table form.
- On the form, locate the **Form Context Menu/Additional Actions** option on the left side of the form header.
- Click the **Form Context Menu/ Additional Actions** icon.
- Click configure, select **Form Layout**.



- The **Form Layout** page will open, where you can configure the fields displayed on the form
- After opening **Form Layout**, click **New** to create a new field.
- Enter the field details:
- **Column Label:** EmployeeID
- **Type:** String
- Click **Add** to create the field.

- Repeat the same steps to create the following fields:
    - **Employee Name** – Type: String
    - **Email** – Type: String
    - **Department** – Type: String
    - **Location** – Type: String
- After creating all the required fields, click **Save**.
- Verify that the newly created fields are displayed on the **Employee Test** form.

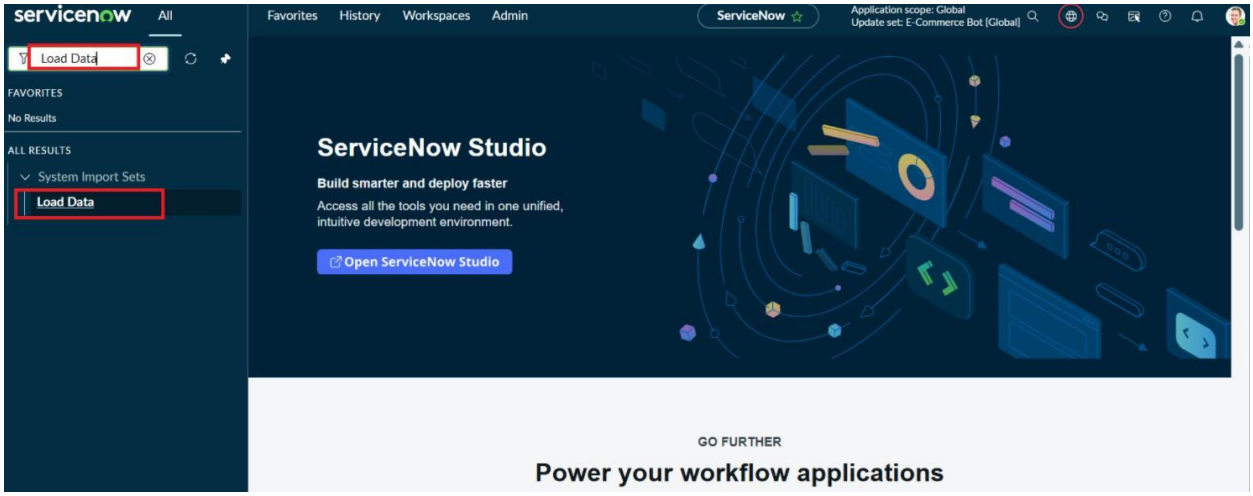


Click on Submit or Save.

Phase 3: Import Set Table

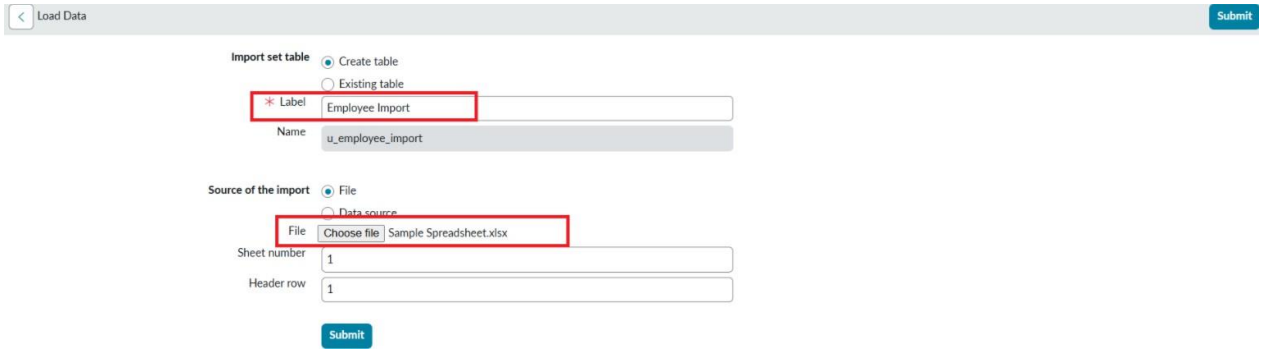
Navigation:

In Application Navigator → Load Data



Click on Load Data and create an import table
Fill details as shown below

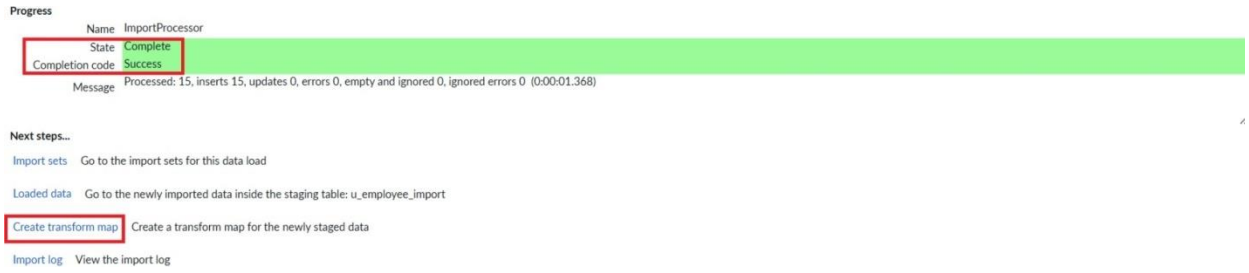
- Label: Employee Import
- Name: u_employee_import(autopopulated)



- Click on Submit

Once we click on Submit, we can see the success message as shown below.

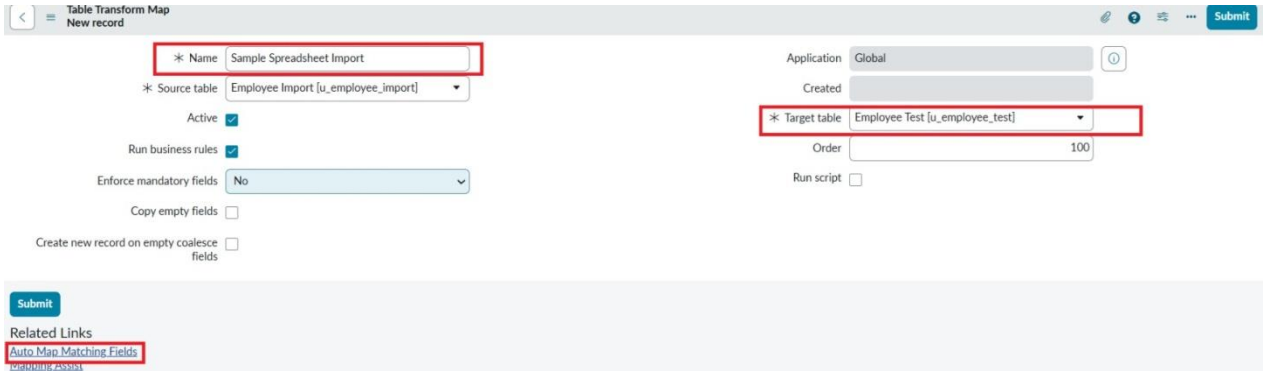
Also we need to make sure to click on Create Transform Map.



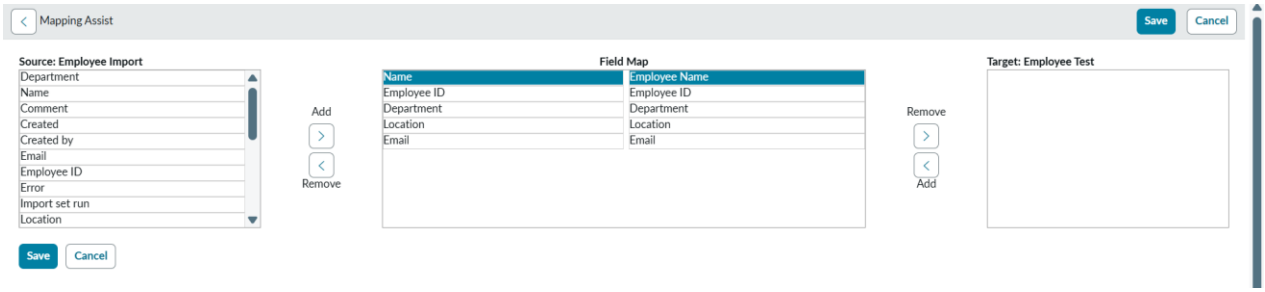
Phase 4: Create Transform Map

In this phase we need to create the Transform Map by click ling on the Create Transform Map buttonas shown in above screenshot.

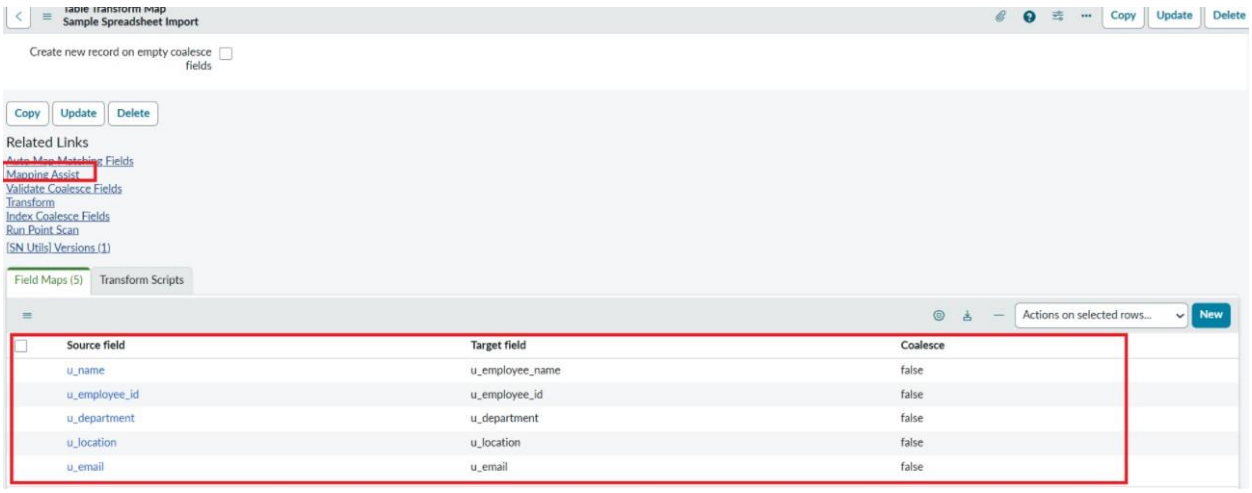
Give inputs as shown below.
Name: Sample Spread sheet
Import
Source Table: Employee Import is auto-populated
Target Table: Employee Test- as we create dearlier
Click on Auto Map Matching fields.



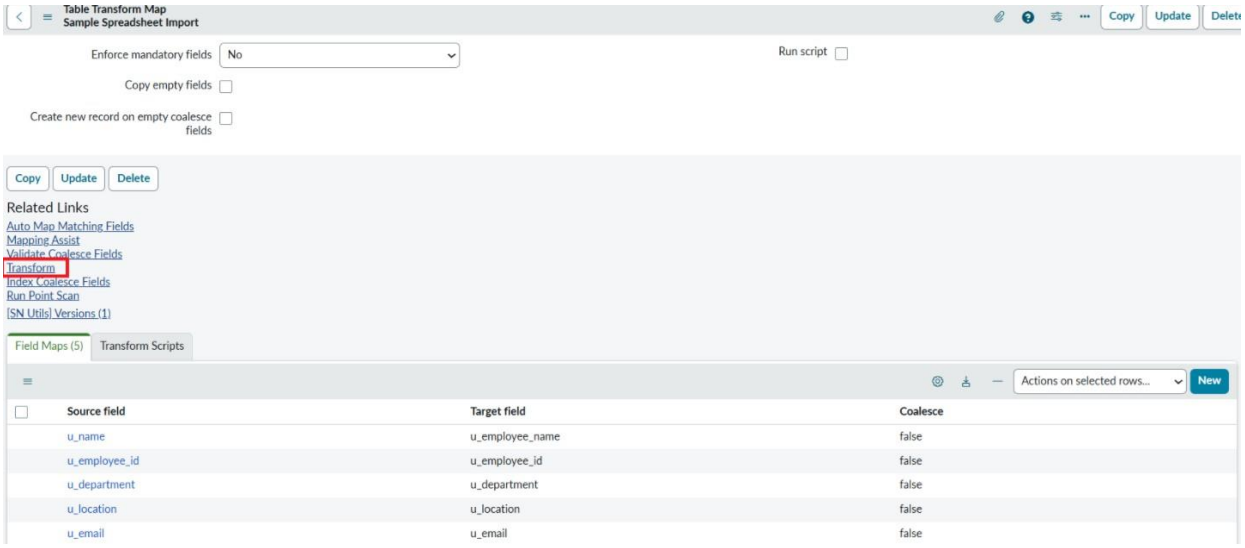
Now click on Mapping Assistant map the fields with Source &Target Table



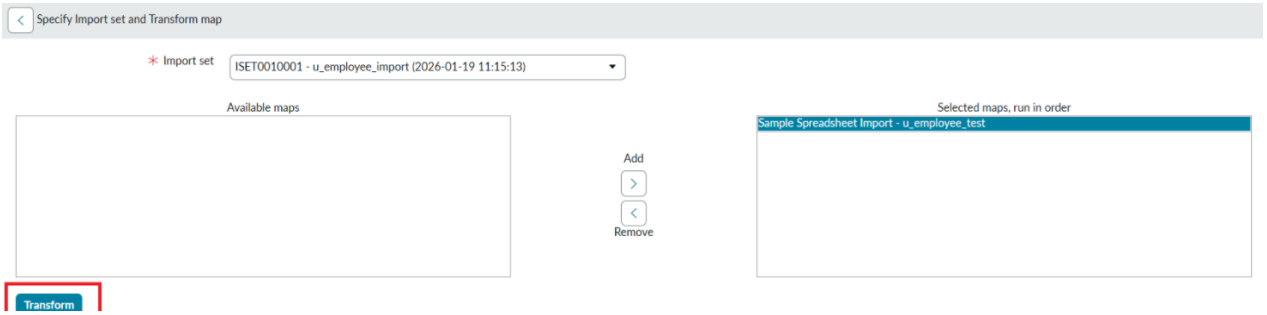
Click on Save
Make sure the both Source &Target table field sarem apped as shown below.



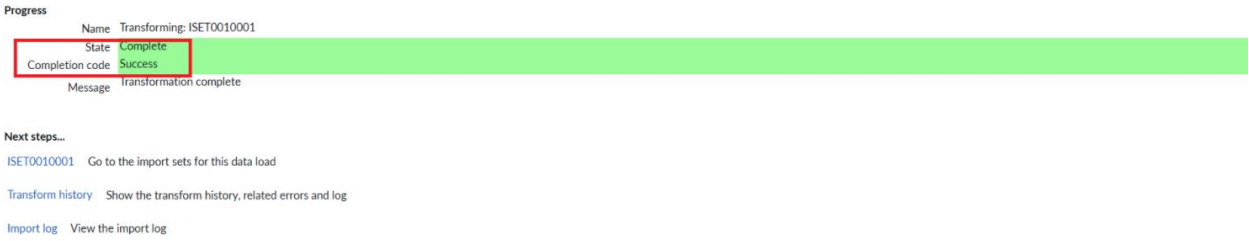
- NowclickonTransformasshownbelow



- NowclickonTransformButtonasshownbelow.



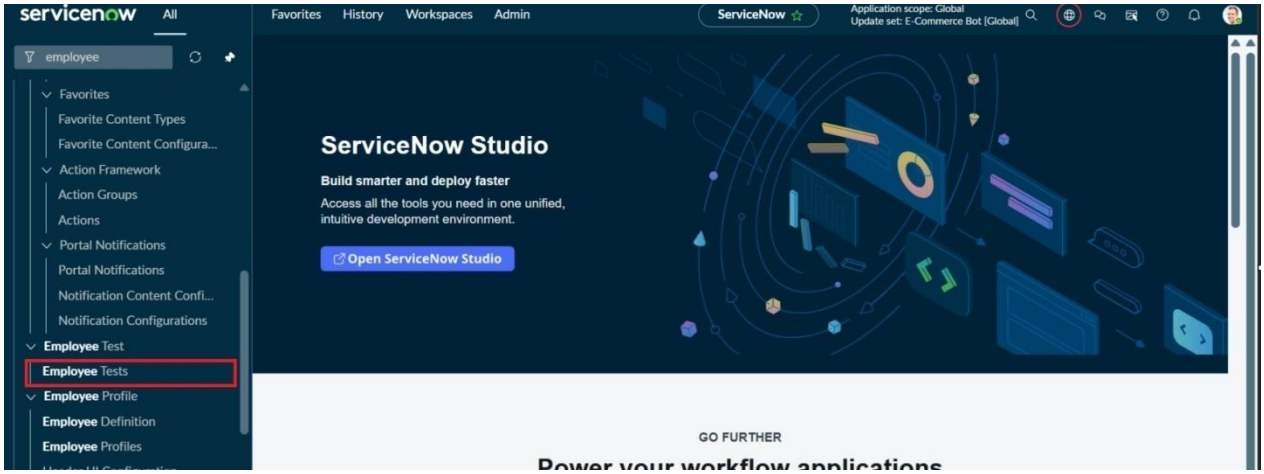
- AfterclickingTransform,makesureyouseethesuccessmessageasshownbelow.



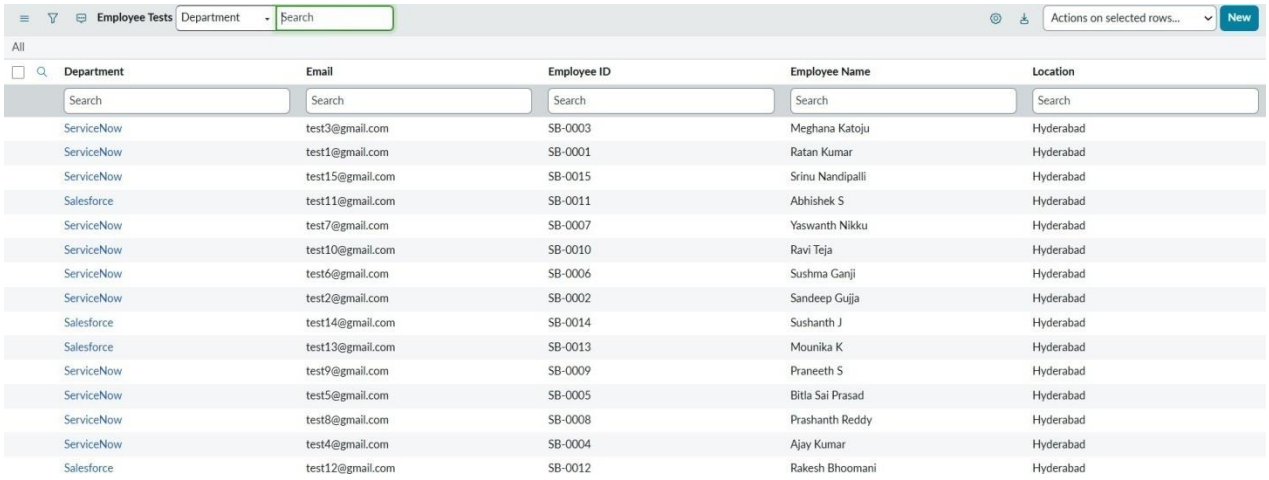
Phase5:TransformData&Validate

Navigation:

- OpenEmployeeTestTablefromapplicationnavigatorasshownbelow.

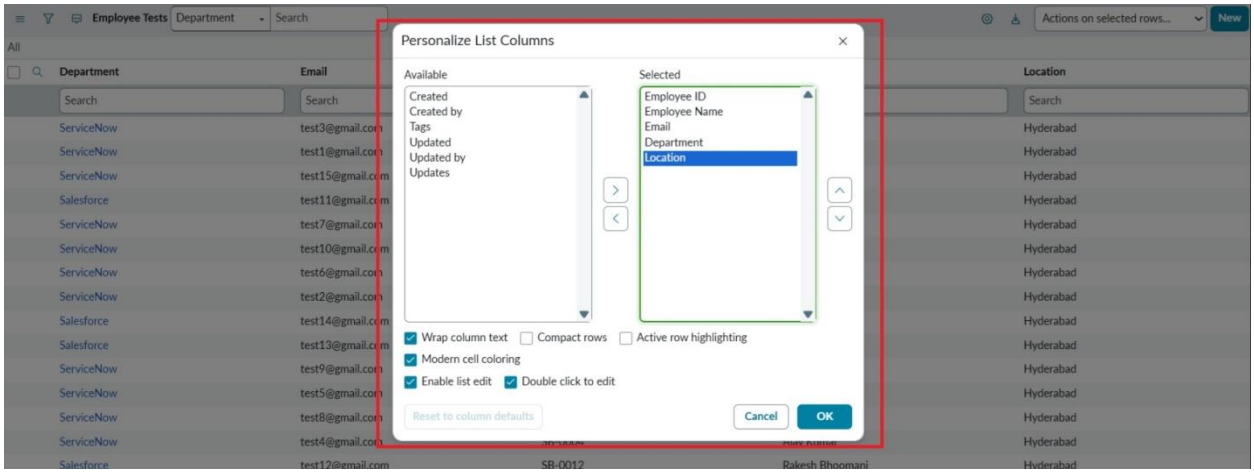


- Asourimportgotsucceeded,alltherecordsarecreatedintheEmployeeTest Table.



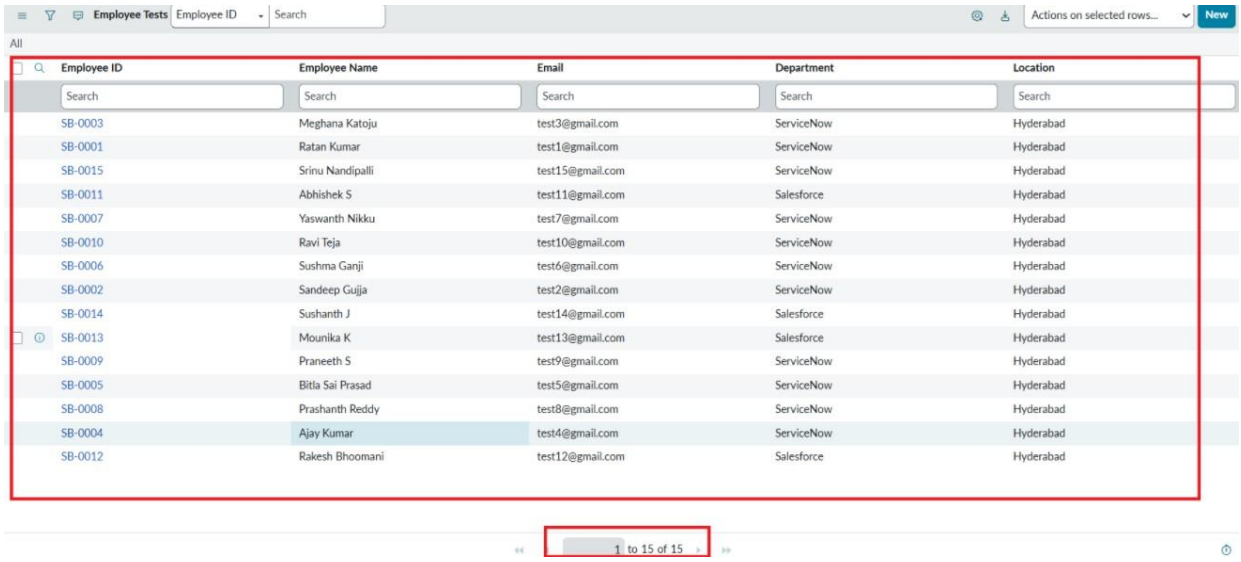
Department	Email	Employee ID	Employee Name	Location
Search	Search	Search	Search	Search
ServiceNow	test3@gmail.com	SB-0003	Meghana Katoju	Hyderabad
ServiceNow	test1@gmail.com	SB-0001	Ratan Kumar	Hyderabad
ServiceNow	test15@gmail.com	SB-0015	Srinu Nandipalli	Hyderabad
Salesforce	test11@gmail.com	SB-0011	Abhishek S	Hyderabad
ServiceNow	test7@gmail.com	SB-0007	Yaswanth Nikku	Hyderabad
ServiceNow	test10@gmail.com	SB-0010	Ravi Teja	Hyderabad
ServiceNow	test6@gmail.com	SB-0006	Sushma Ganji	Hyderabad
ServiceNow	test2@gmail.com	SB-0002	Sandeep Gujja	Hyderabad
Salesforce	test14@gmail.com	SB-0014	Sushanth J	Hyderabad
Salesforce	test13@gmail.com	SB-0013	Mounika K	Hyderabad
ServiceNow	test9@gmail.com	SB-0009	Praneeth S	Hyderabad
ServiceNow	test5@gmail.com	SB-0005	Bitla Sai Prasad	Hyderabad
ServiceNow	test8@gmail.com	SB-0008	Prashanth Reddy	Hyderabad
ServiceNow	test4@gmail.com	SB-0004	Ajay Kumar	Hyderabad
Salesforce	test12@gmail.com	SB-0012	Rakesh Bhoomani	Hyderabad

- MakesureyourearrangethefieldsaccordinglybyusingPersonalisedList Columns



ClickonOK

FinalResult:
AlltherecordsaresuccessfullyimportedintotheServiceNowsystemwithImportSet, Transform Map.

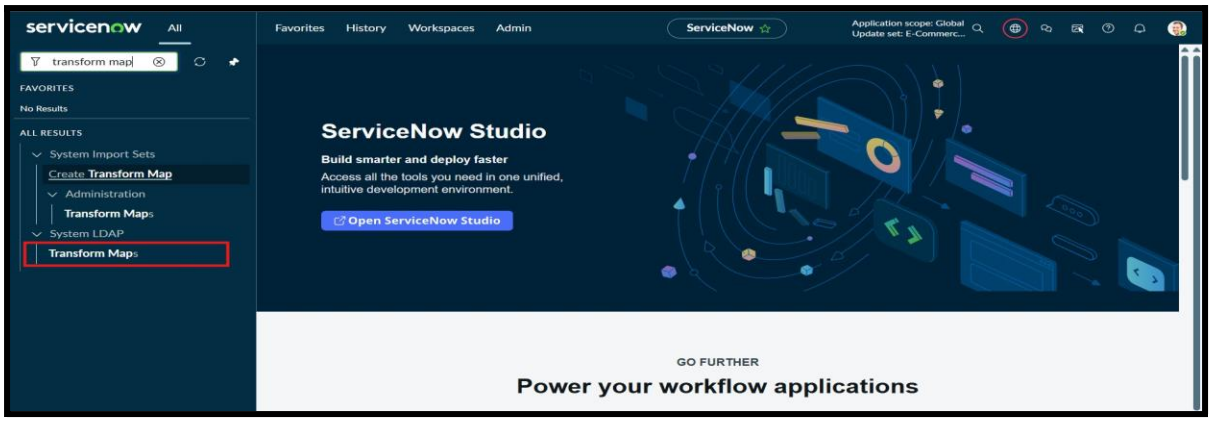


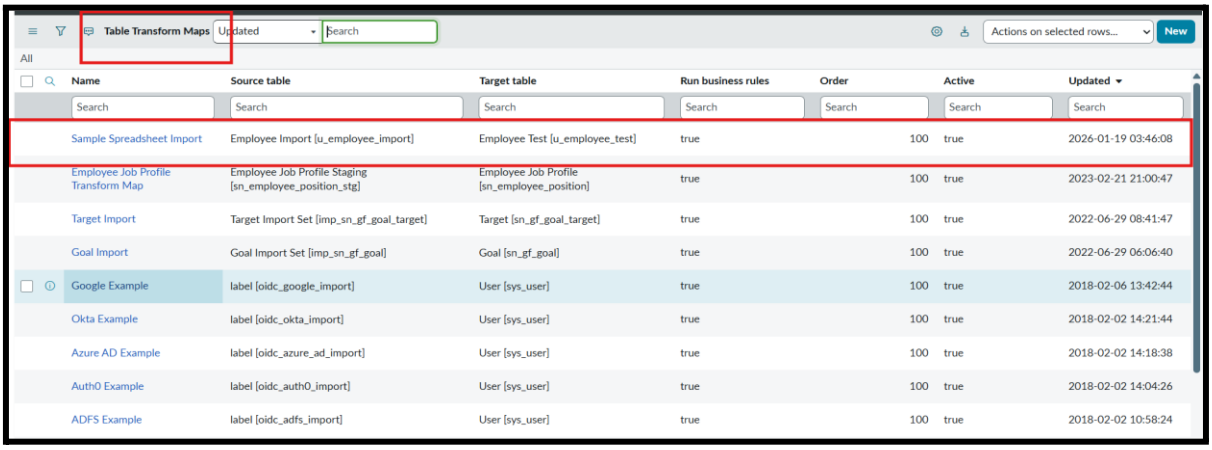
Phase6:EnableCoalescetoAvoidDuplicateRecords

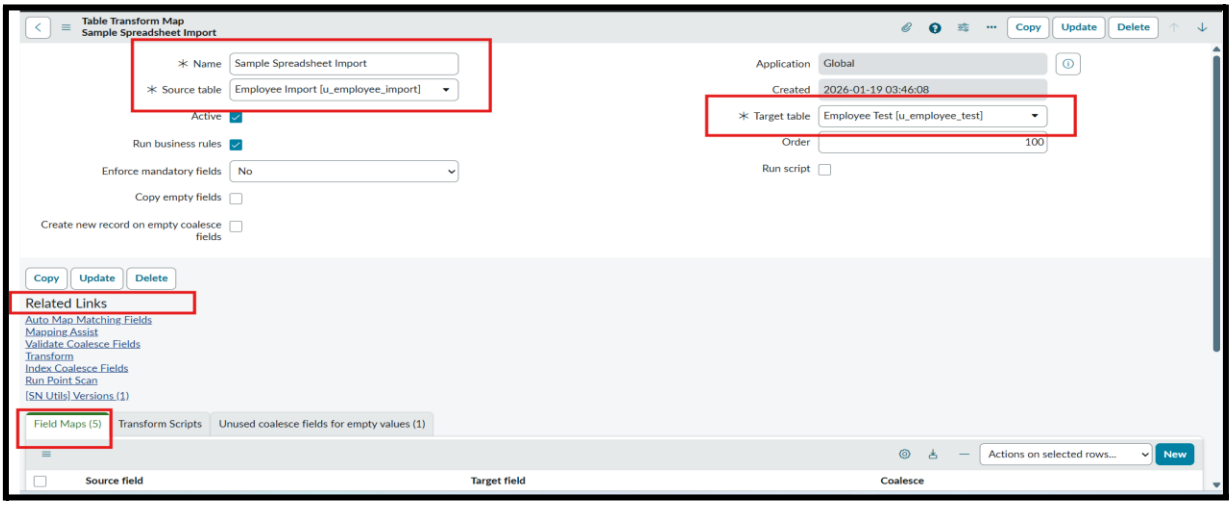
- Inrealprojects,thesameemployeedatamaybeimportedmultipletimes.
- Topreventduplicateemployeerecords,weuse**Coalesce**.
- TheCoalescewillupdatethesamerecordswithnewvaluesifweaddandthe duplicate values are not inserted.

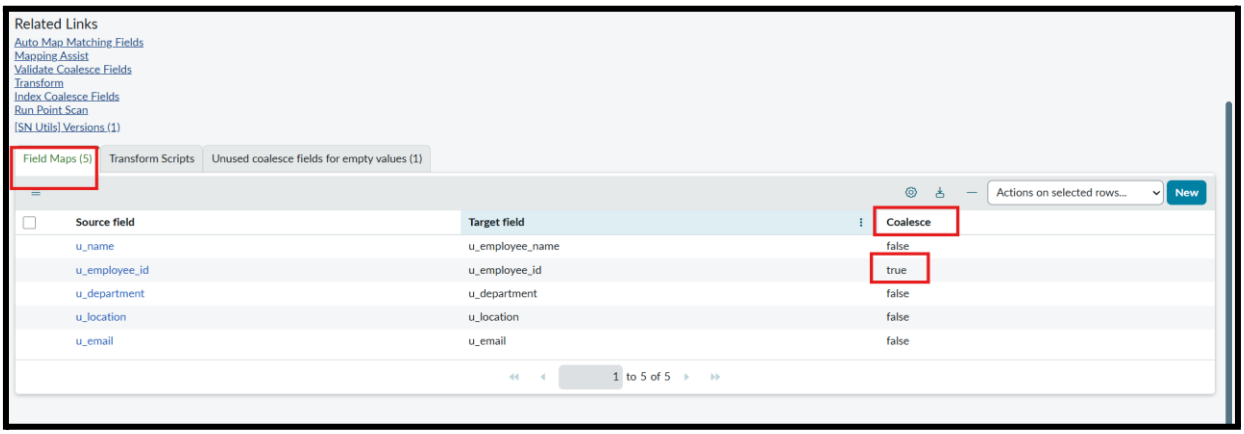
Steps:

1. OpenTransformmapUnderSystemLDAPOnAllmenus.
2. OpentheCreatedTransformmap,NameisSampleSpreadsheetImport.
3. UnderRelatedlist,selecttheFieldmapsSectionandMakeanyonefieldCoalesce True which appears in False.
4. Nowsavetheform.
5. Thenagainaddsomedatawhichisin Excelformat.


servicenow
All
Favorites
History
Workspaces
Admin
ServiceNow
transform map
FAVORITES
No Results
ALL RESULTS
System Import Sets
Create Transform Map
Administration
Transform Maps
System LDAP
Transform Maps
ServiceNow Studio
Build smarter and deploy faster
Access all the tools you need in one unified, intuitive development environment.
Open ServiceNow Studio
GO FURTHER
Power your workflow applications


Table Transform Maps
Updated
Search
Actions on selected rows...
New
All
Name
Source table
Target table
Run business rules
Order
Active
Updated
Sample Spreadsheet Import
Employee Import [u_employee_import]
Employee Test [u_employee_test]
true
100
true
2026-01-19 03:46:08
Employee Job Profile Transform Map
Employee Job Profile Staging [sn_employee_position_stg]
Employee Job Profile [sn_employee_position]
2023-02-21 21:00:47
Target Import
Target Import Set [imp_sn_gf_goal_target]
Target [sn_gf_goal_target]
2022-06-29 08:41:47
Goal Import
Goal Import Set [imp_sn_gf_goal]
Goal [sn_gf_goal]
2022-06-29 06:06:40
Google Example
label [oidc_google_import]
User [sys_user]
2018-02-06 13:42:44
Okta Example
label [oidc_okta_import]
2018-02-02 14:21:44
Azure AD Example
label [oidc_azure_ad_import]
2018-02-02 14:18:38
Auth0 Example
label [oidc_auth0_import]
2018-02-02 14:04:26
ADFS Example
label [oidc_adfs_import]
2018-02-02 10:58:24

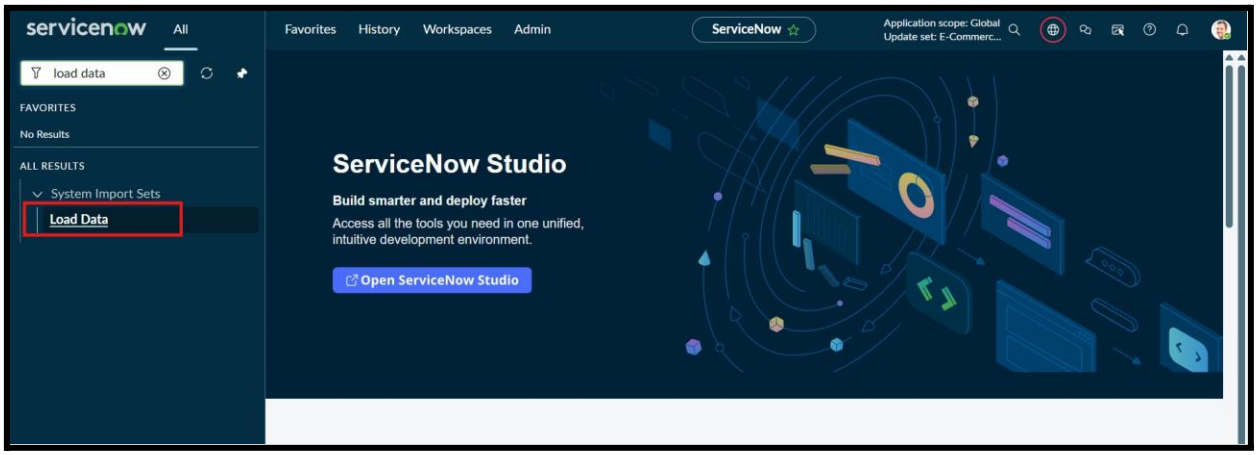

Table Transform Map
Sample Spreadsheet Import
Copy
Update
Delete
Name
Sample Spreadsheet Import
Source table
Employee Import [u_employee_import]
Active
Run business rules
Enforce mandatory fields
No
Copy empty fields
Create new record on empty coalesce fields
Application
Global
Created
2026-01-19 03:46:08
Target table
Employee Test [u_employee_test]
Order
100
Run script
Related Links
Auto Map Matching Fields
Mapping Assist
Validate Coalesce Fields
Transform
Index Coalesce Fields
Run Point Scan
[SN Utils] Versions (1)
Field Maps (5)
Transform Scripts
Unused coalesce fields for empty values (1)
Actions on selected rows...
New
Source field
Target field
Coalesce

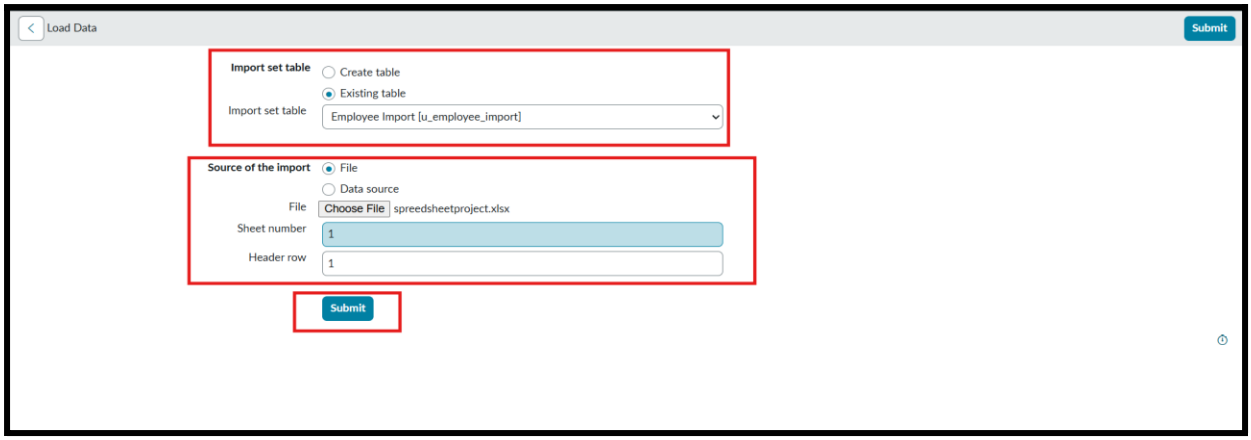


PHASE7 : Inserting New Data In Excel Format

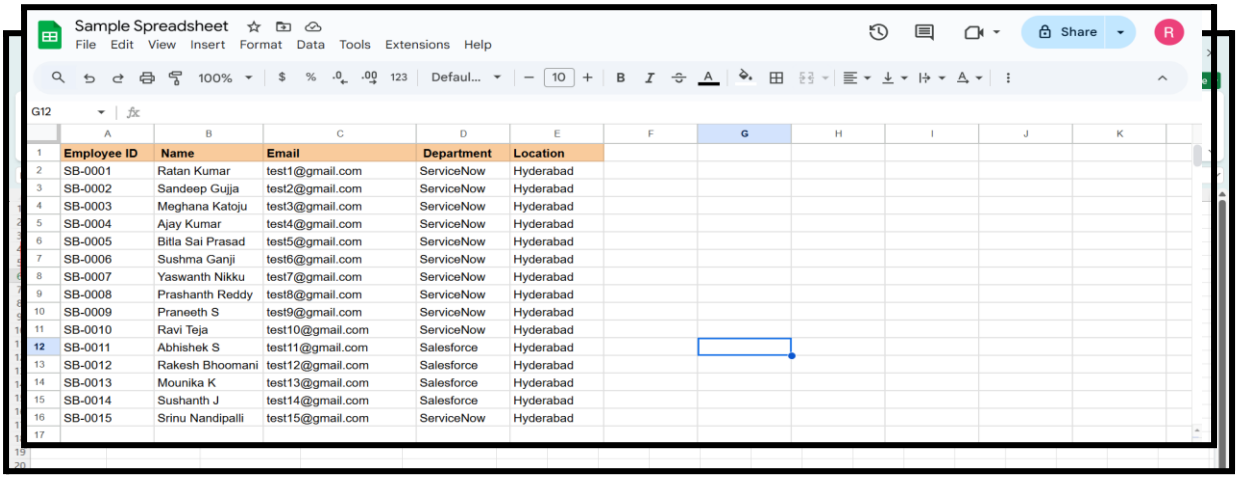
Steps:

- NavigatetoAllmenus.
- SearchforLoaddataUnderSystemImportsetOpenLoadData.
- Fillthedetailsas
- ImportsettableasselectExistingtable{EmployeeImport}.
- SourceofImportisSelectFileandchoosethefilefromFiles.
- Sheetnumber:1,Headerrow:1.
- NowClicksubmit.



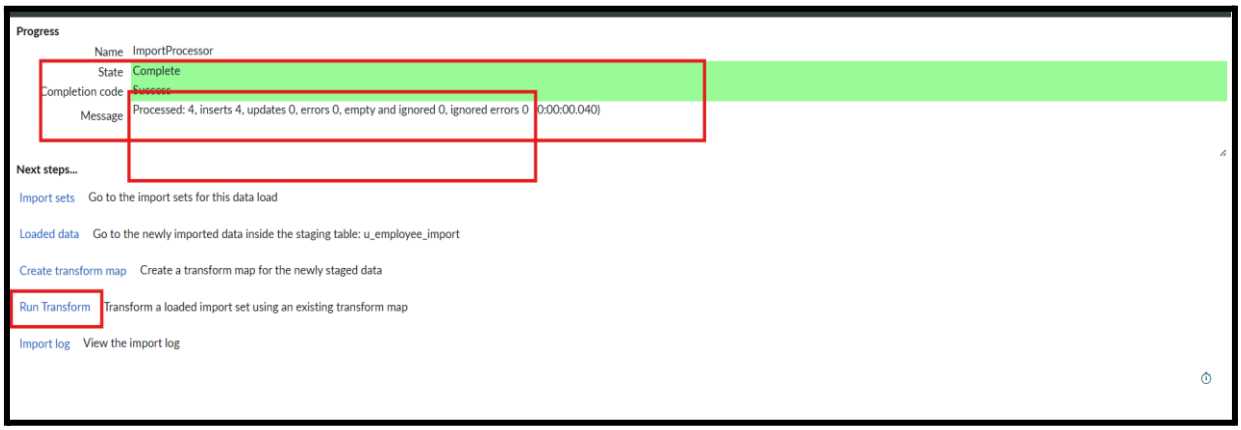


- NewlyCreatedExcelDataisthis.

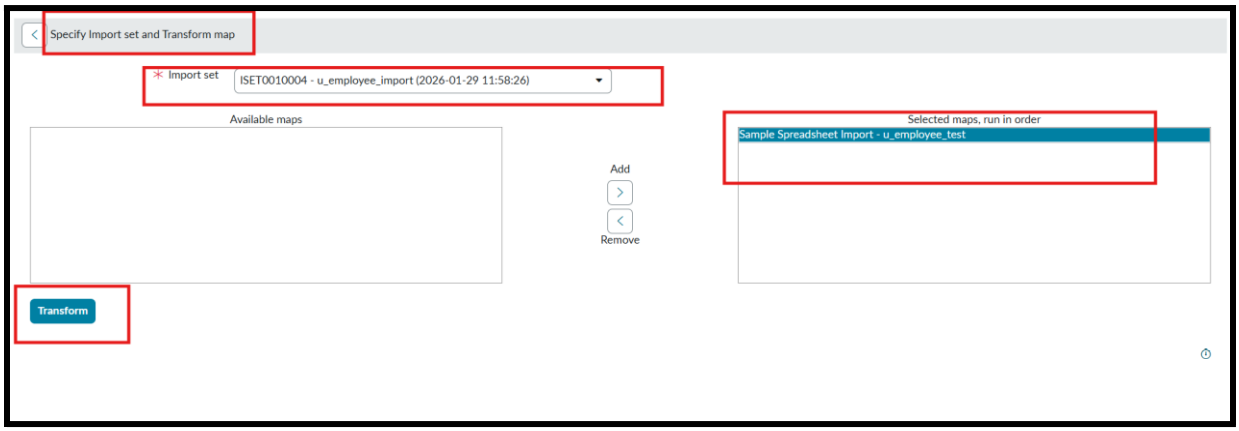


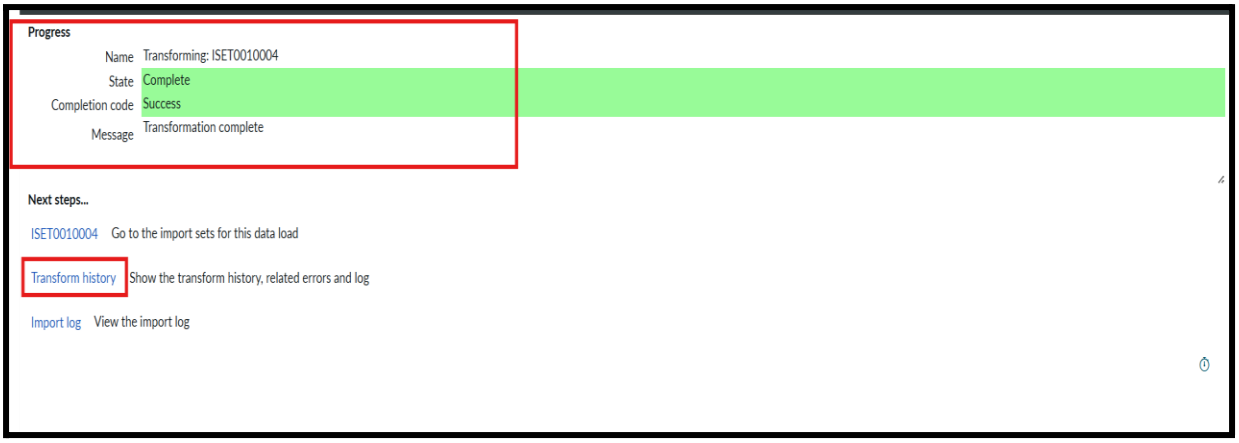
Employee ID	Name	Email	Department	Location
SB-0001	Ratan Kumar	test1@gmail.com	ServiceNow	Hyderabad
SB-0002	Sandeep Gujja	test2@gmail.com	ServiceNow	Hyderabad
SB-0003	Meghana Katoju	test3@gmail.com	ServiceNow	Hyderabad
SB-0004	Ajay Kumar	test4@gmail.com	ServiceNow	Hyderabad
SB-0005	Bitla Sai Prasad	test5@gmail.com	ServiceNow	Hyderabad
SB-0006	Sushma Ganji	test6@gmail.com	ServiceNow	Hyderabad
SB-0007	Yaswanth Nikku	test7@gmail.com	ServiceNow	Hyderabad
SB-0008	Prashanth Reddy	test8@gmail.com	ServiceNow	Hyderabad
SB-0009	Praneeth S	test9@gmail.com	ServiceNow	Hyderabad
SB-0010	Ravi Teja	test10@gmail.com	ServiceNow	Hyderabad
SB-0011	Abhishek S	test11@gmail.com	Salesforce	Hyderabad
SB-0012	Rakesh Bhoomani	test12@gmail.com	Salesforce	Hyderabad
SB-0013	Mounika K	test13@gmail.com	Salesforce	Hyderabad
SB-0014	Sushanth J	test14@gmail.com	Salesforce	Hyderabad
SB-0015	Srinu Nandipalli	test15@gmail.com	ServiceNow	Hyderabad

- Itisthepreviouslyaddeddata.
- Now,Inthisnewdataweadded4rowsofthe2rowswiththesameexistingdata with different Name changed and Email changed.
- ThatRavitejaEmployeeEmailwasChangedto [test10@gmail.com→ test18@gmail.com].
- AndEmployeeIDwithSB-OOO4namewaschangedto [Ajay kumar → Ajay].
- AndtwoothernewEmployeeID’sareadded.
- NowaftersubmittingtheformitappearsasStateCompletedandSuccess.

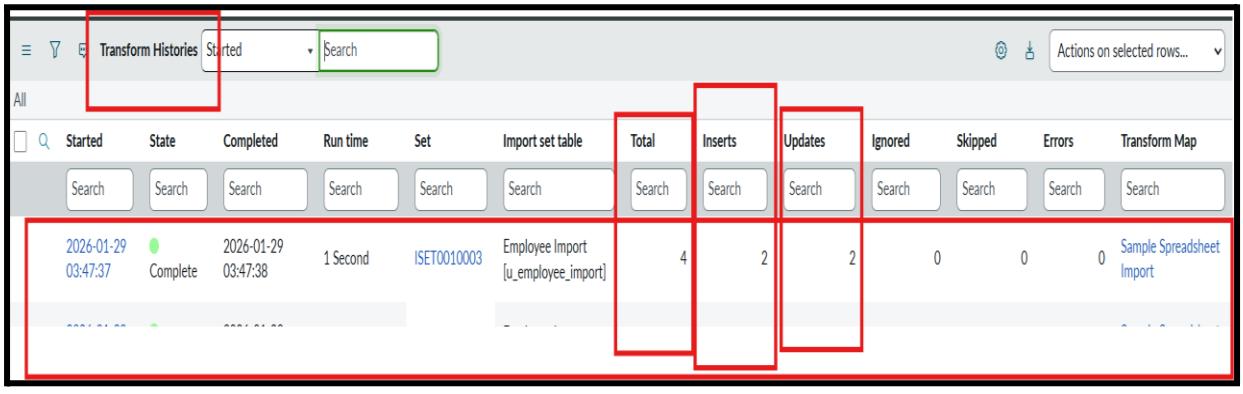


- NowClicktheRuntransformUnderthenextsteps.
- ThenitopensSpecifyImportsetandtransformmapwhichispreviouslyselected transform map then Click Transform button.

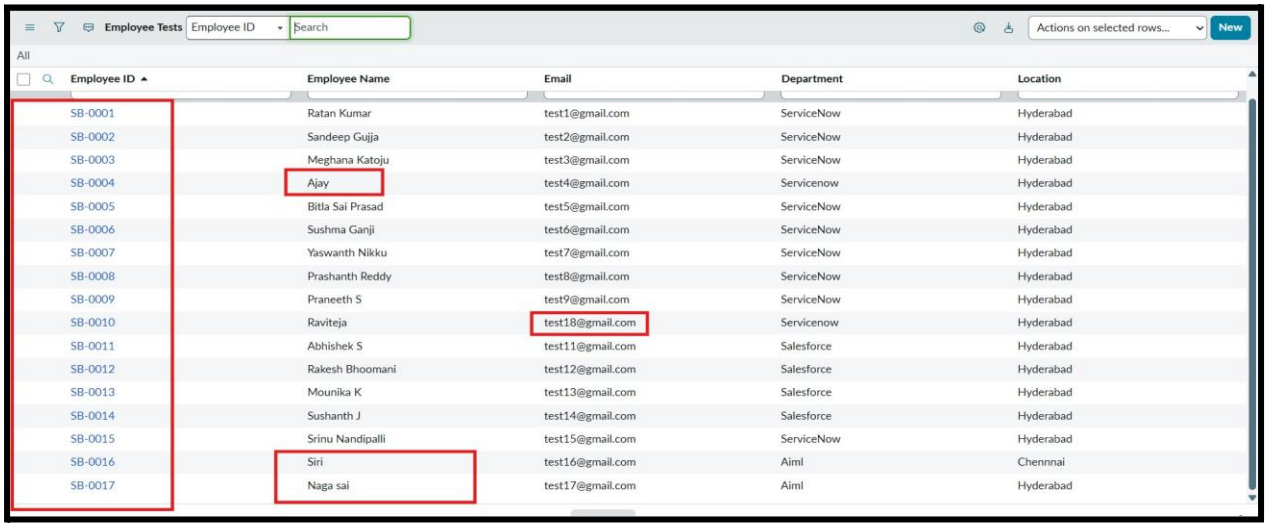




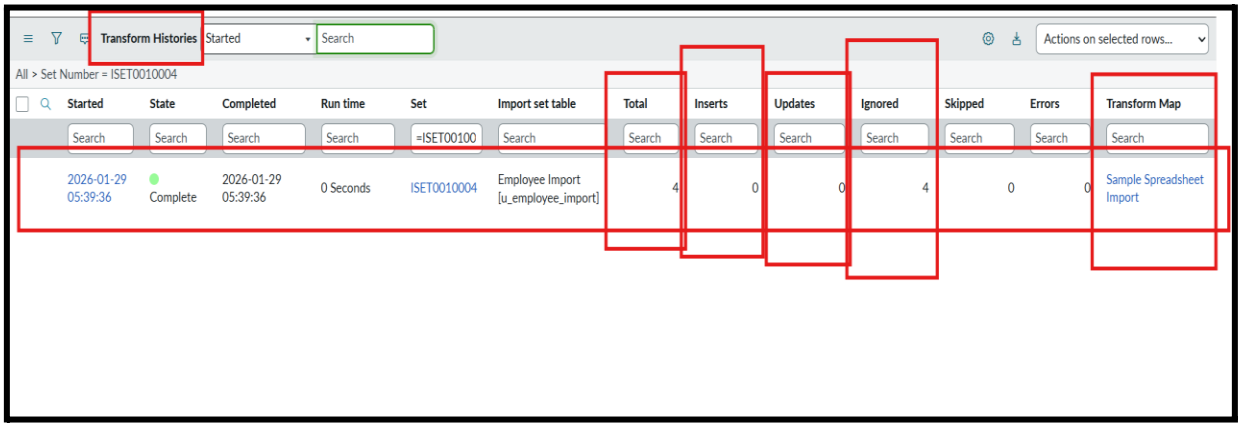
- NowtheStateiscompletedandSuccess.
- NowagainclickontheTransformhistorythenitshowsthenewlyaddeddataand its status of Inserted and Skipped values out the given data.



- Fromtheabovefigure:total4rowsareuploadedoutof2areInsertedthataretwo new ID's.
- And2rowsareupdatesthatareSameID'swithdifferentNameanddifferent Email that we changed.
- ThenOpentheEmployeeTeststableandseetherowsupdatedandinserted.



- NowRepeatthePhase7withthesamestepsandsameExcelsheetandSubmitto check how many rows are updated and Inserted with the same data.
- ThenOpentheTransformhistorytocheck.



- Herewecanseethetotal4areuploaded.
- Onthat0rowsareInsertedandupdates0andIgnored4rows.
- Thatsamesheetweuploadedtothesametransformmapwecreated.

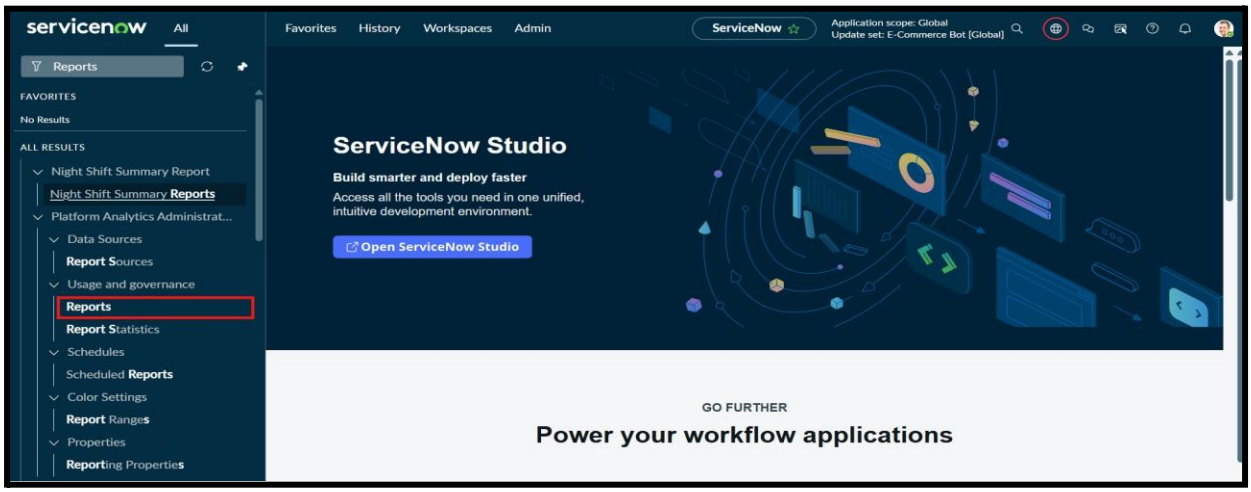
PHASE8:CreateReportsonEmployeeTestTable

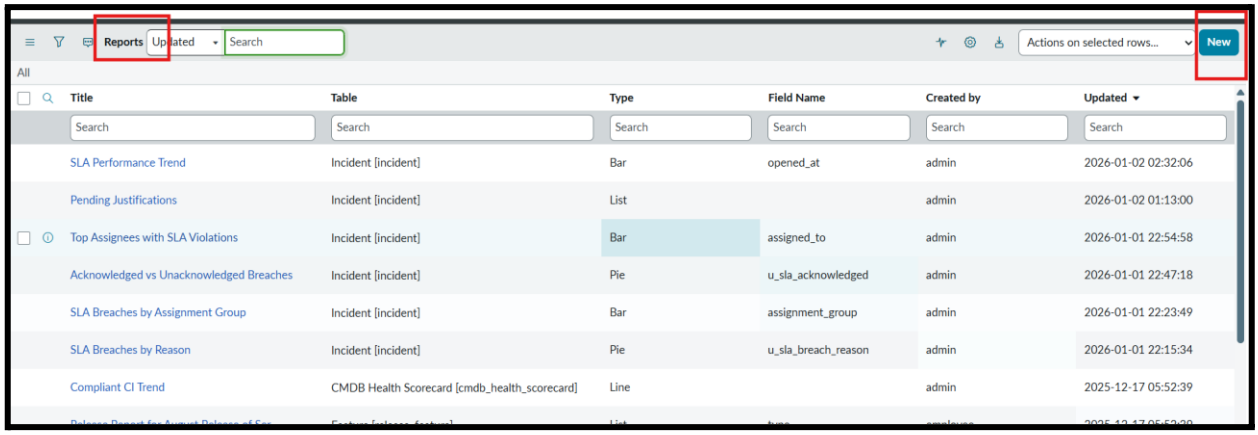
NavigationSteps:

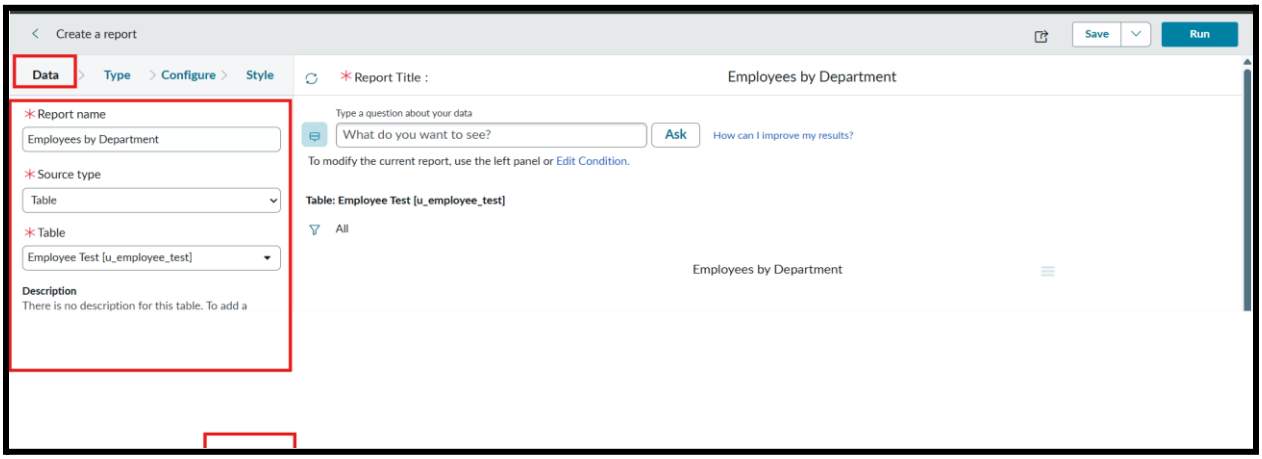
- OpenAllmenus,Searchfor**Reports**.
- Under**Usageandgovernance**SelectReports.
- Clickthe**New**buttontocreateanewreport.
- Create**3Report**sasfollows.

Report1:EmployeesbyDepartment

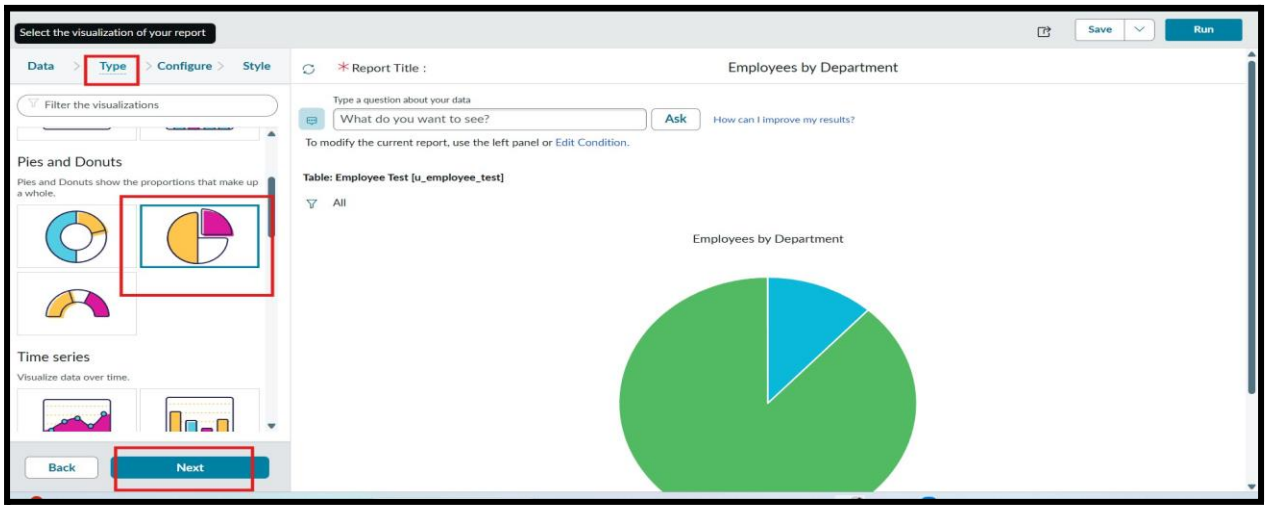
- InData
- Reportname:**EmployeesbyDepartment**.
- SourceType:**Table**.
- Table:**EmployeeTest**,NowclicktheNextbutton.



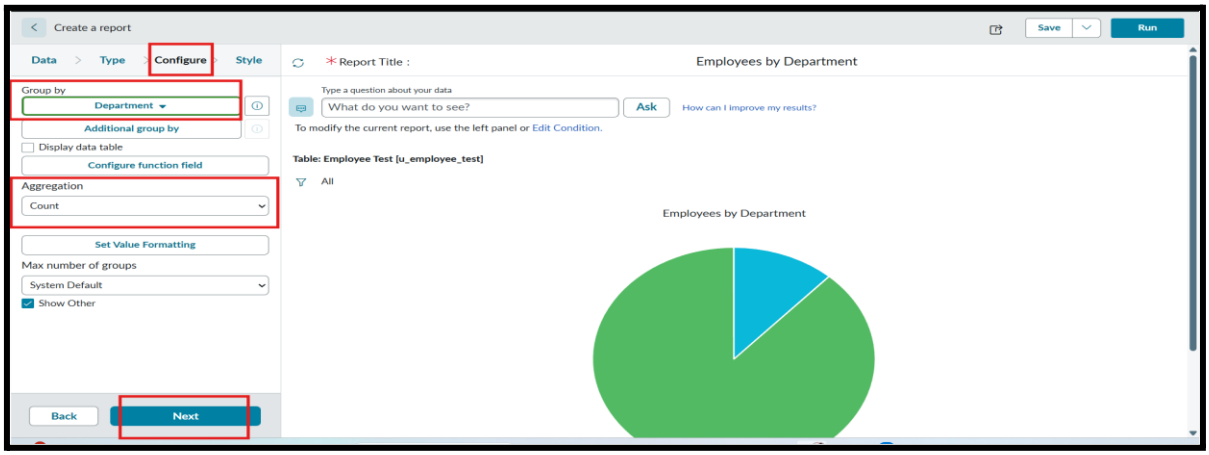




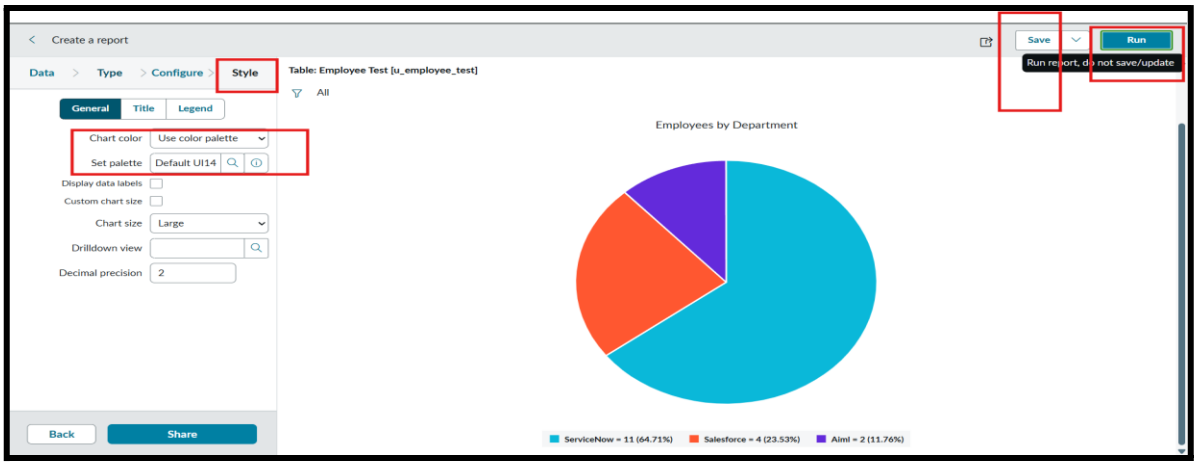
- NowafterClickingnextintypeSelect**Piechart**.
- Andclickthe**next**button.



- NowInConfigureSelectGroupby**Department**Field.
- IntheAggregationFieldSelect**Count.**
- ThenClick**next**.



- NowInStyleLeavetheChartcolorby**default.**
- AndClickthe**Run**buttontoExecutethegivendataReport.
- AndClick**save**buttontoSavetheReport.



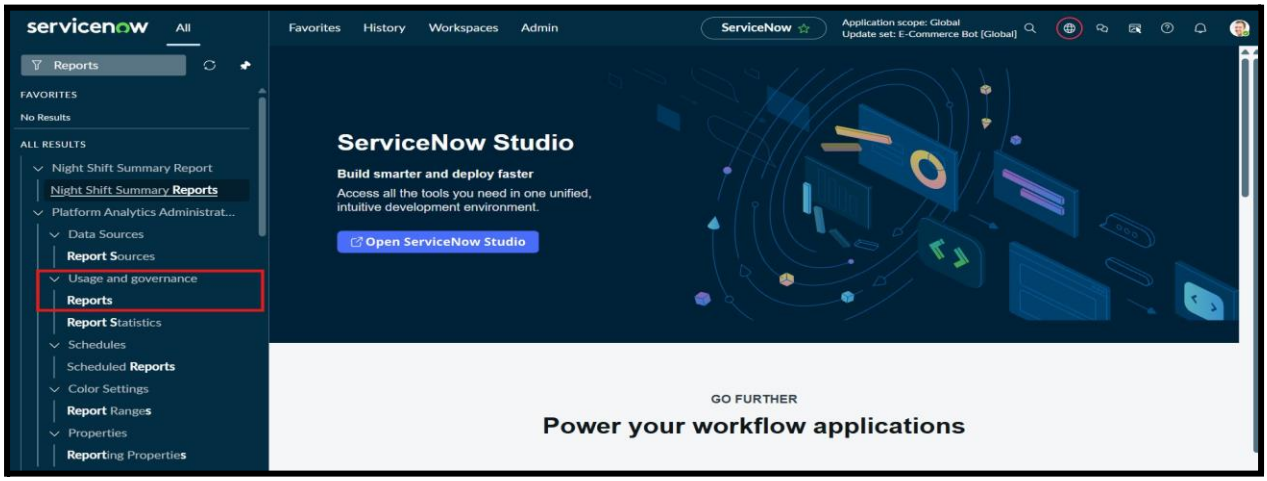
Report2:EmployeesbyLocation

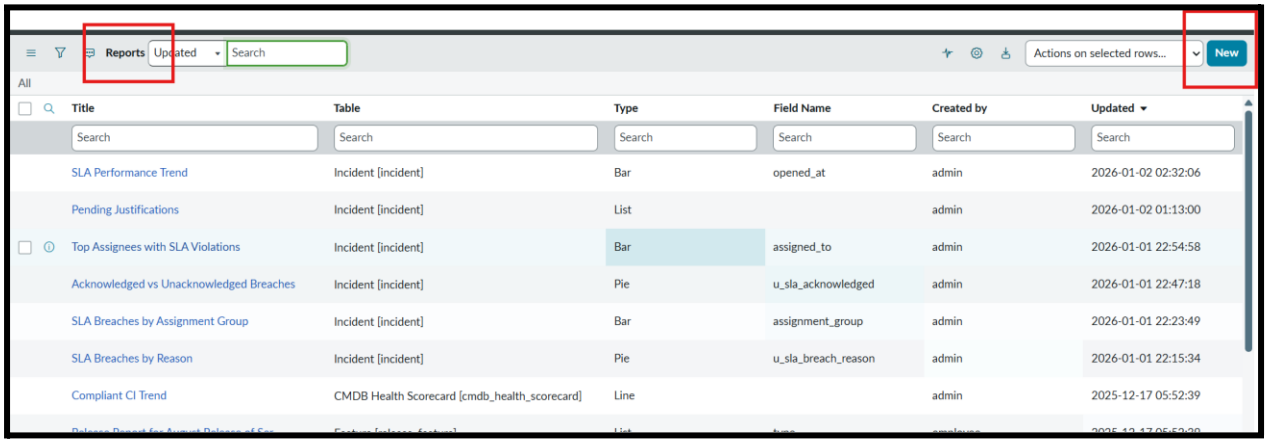
Configuration

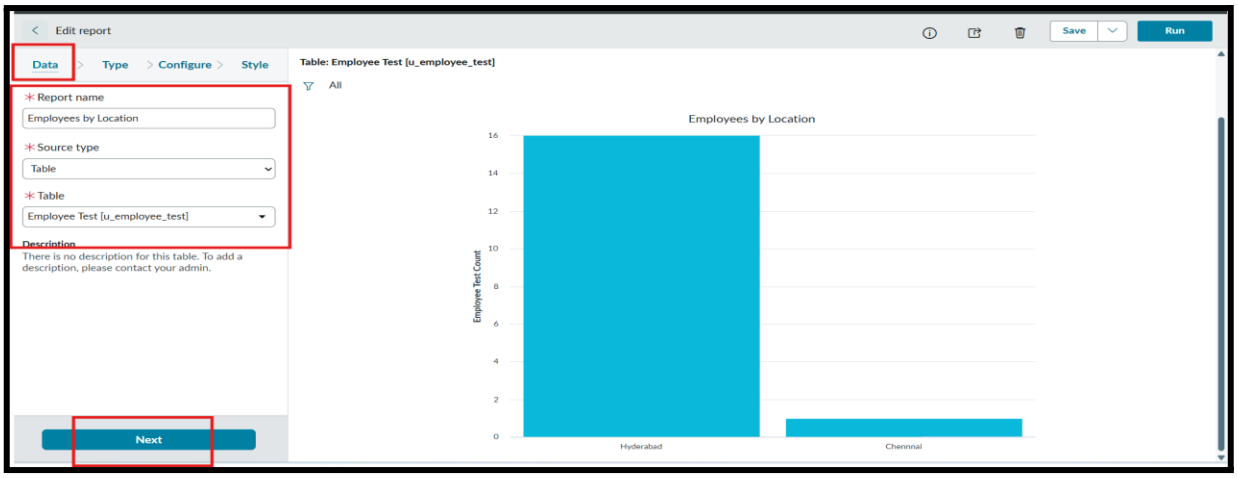
- Table:EmployeeTest
- Type:**BarChart**
- Aggregation:**Count**
- GroupBy:**Location**

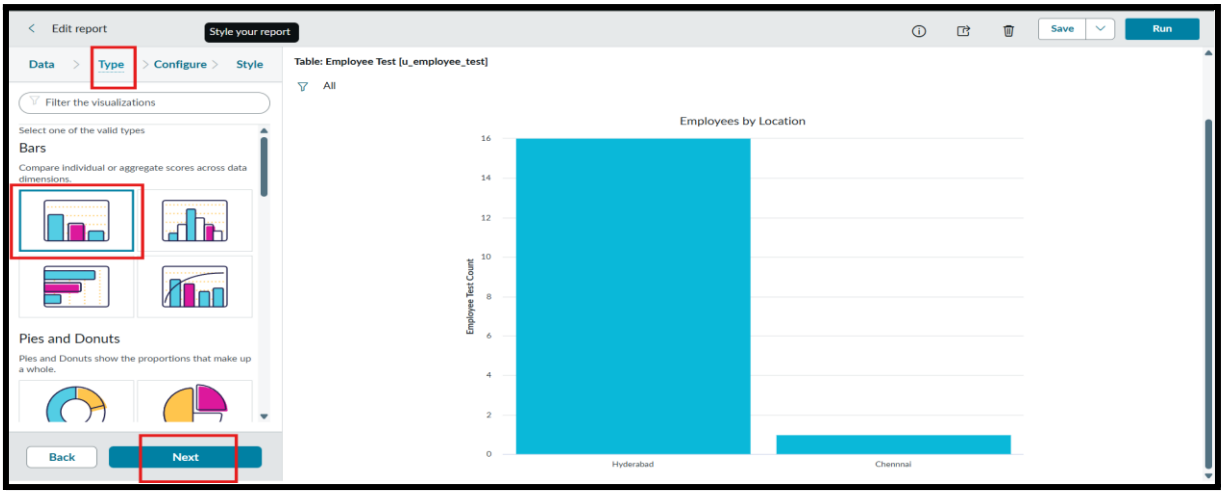
Steps:

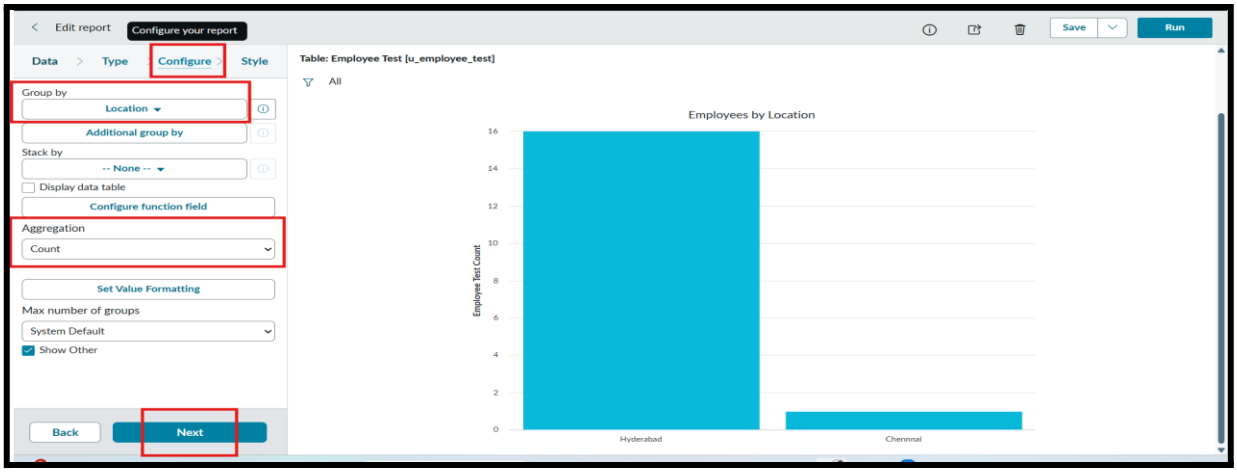
1. CreateaNewReport.
2. Reportname:**EmployeesbyLocation.**
3. TableSelect:**EmployeeTest**.
4. SelectType:**BarChart.**
5. InConfigureSelectGroupbyField:**Location.**
6. AndAggregationis**Count**.
7. NowClick**Run**toExecutetheReportandClickthe**Save**tosavetheReport.

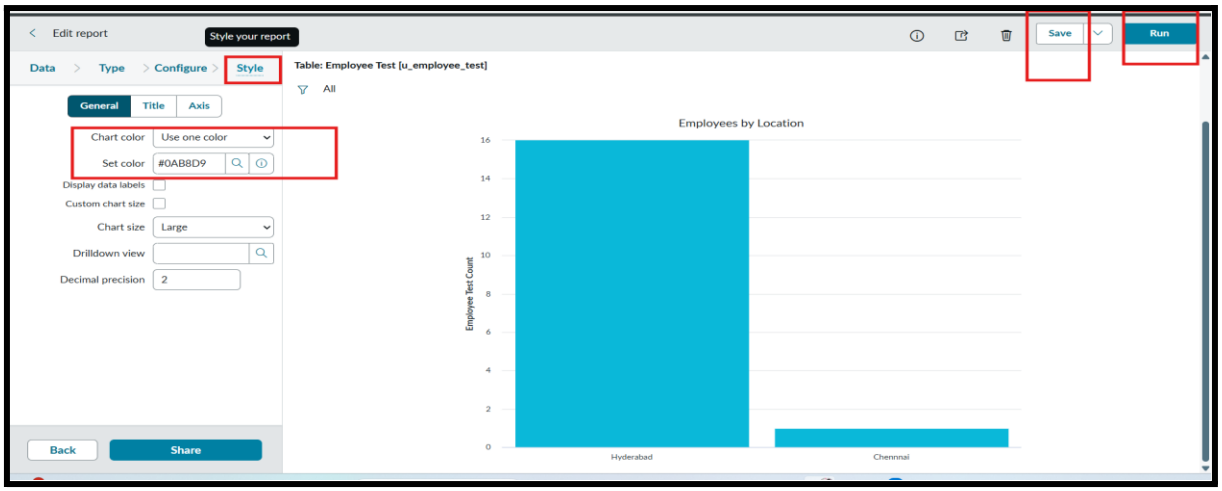





Edit report
Save
Run
Data
Type
Configure
Style
Report name
Employees by Location
Source type
Table
Table
Employee Test [u_employee_test]
Description
There is no description for this table. To add a description, please contact your admin.
Next
Table: Employee Test [u_employee_test]
All
Employees by Location
Employee Test Count
Hyderabad
Chennnai


Edit report
Style your report
Save
Run
Data
Type
Configure
Style
Filter the visualizations
Select one of the valid types
Bars
Compare individual or aggregate scores across data dimensions.
Pies and Donuts
Pies and Donuts show the proportions that make up a whole.
Back
Next
Table: Employee Test [u_employee_test]
All
Employees by Location
Employee Test Count
Hyderabad
Chennnai

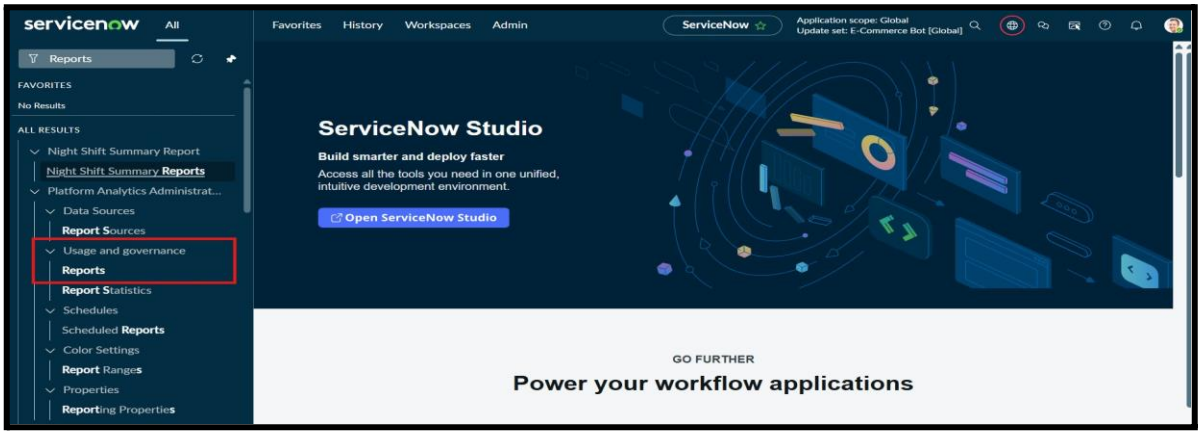

Edit report
Configure your report
Save
Run
Data
Type
Configure
Style
Group by
Location
Additional group by
Stack by
-- None --
Display data table
Configure function field
Aggregation
Count
Set Value Formatting
Max number of groups
System Default
Show Other
Back
Next
Table: Employee Test [u_employee_test]
All
Employees by Location
Employee Test Count
Hyderabad
Chennnai

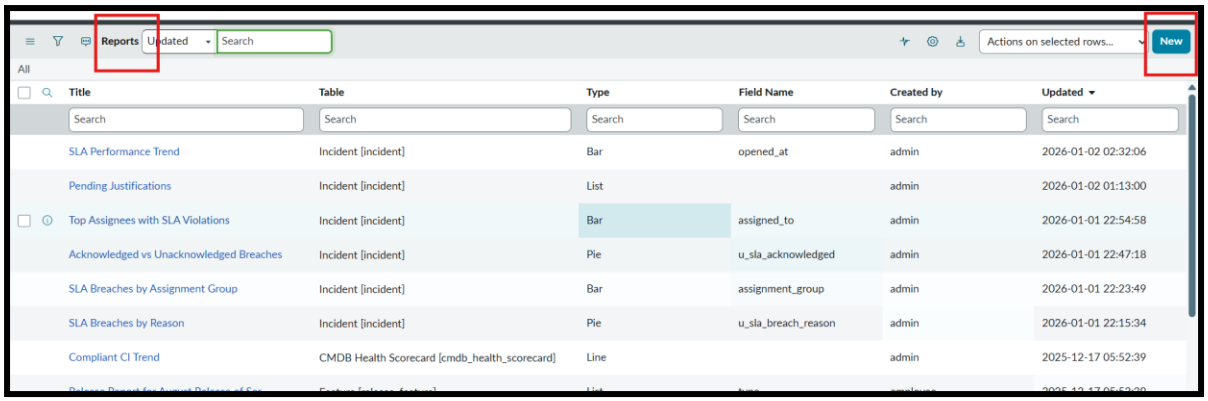


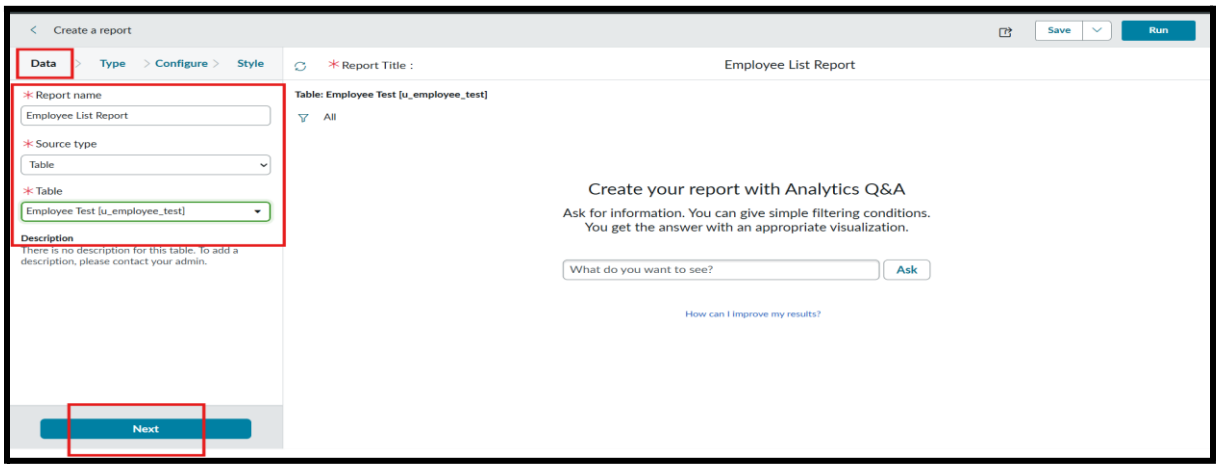
Report3:EmployeeListReport.

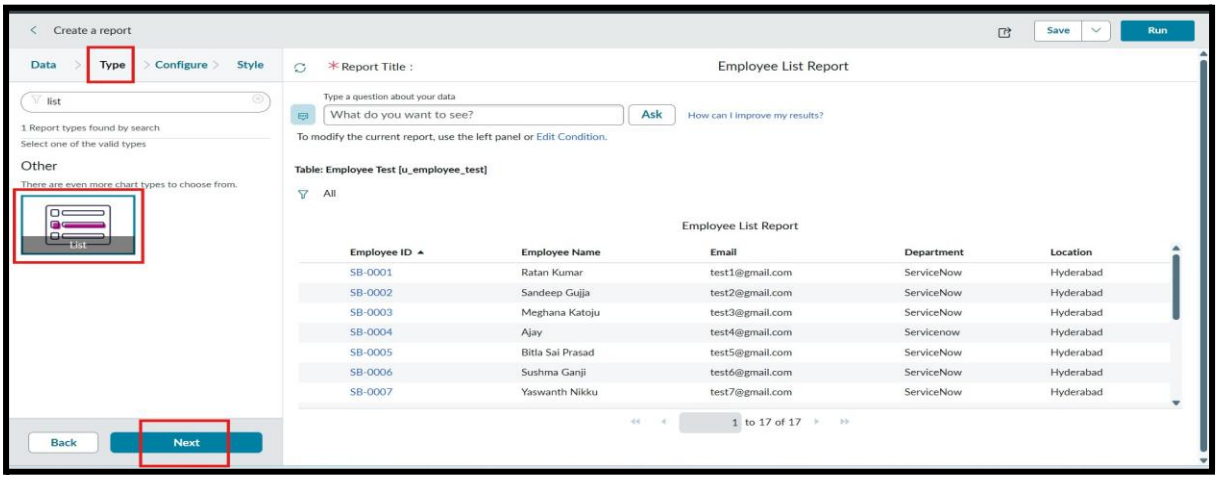
Configuration

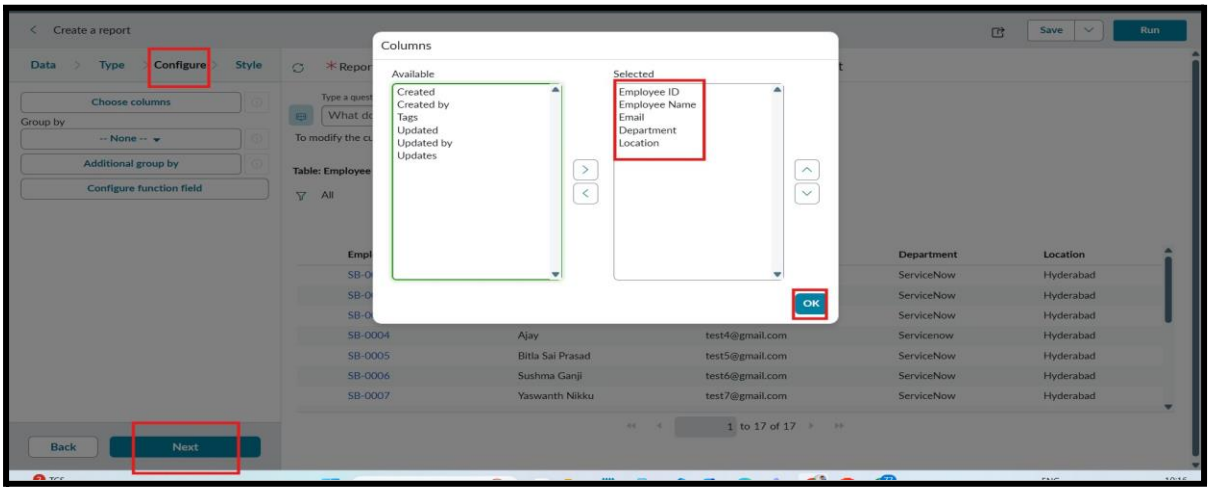
- CreateaNewReportname:**EmployeeListReport.**
- Table:**EmployeeTest.**
- SelectType:**List.**
- InConfigureSelectColumns:
  - **EmployeeID.**
  - **EmployeeName.**
  - **Email.**
  - **Department.**
  - **Location.**
- NowSelect**next**andnorequiredtochooseanythinginStyle.
- NowClick**Run**toExecutetheReportandClick**Save**toSavetheReport.

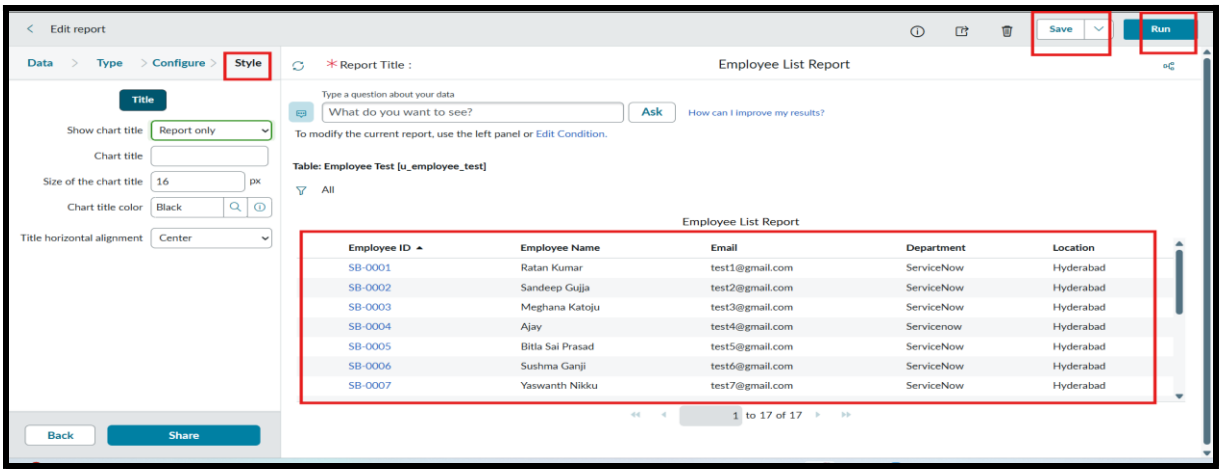



Reports
Updated
Search
Actions on selected rows...
New
All
Title
Table
Type
Field Name
Created by
Updated
SLA Performance Trend
Incident [incident]
Bar
opened_at
admin
2026-01-02 02:32:06
Pending Justifications
Incident [incident]
List
admin
2026-01-02 01:13:00
Top Assignees with SLA Violations
Incident [incident]
Bar
assigned_to
admin
2026-01-01 22:54:58
Acknowledged vs Unacknowledged Breaches
Incident [incident]
Pie
u_sla_acknowledged
admin
2026-01-01 22:47:18
SLA Breaches by Assignment Group
Incident [incident]
Bar
assignment_group
admin
2026-01-01 22:23:49
SLA Breaches by Reason
Incident [incident]
Pie
u_sla_breach_reason
admin
2026-01-01 22:15:34
Compliant CI Trend
CMDB Health Scorecard [cmdb_health_scorecard]
Line
admin
2025-12-17 05:52:39


Create a report
Save
Run
Data
Type
Configure
Style
Report Title :
Employee List Report
Report name
Employee List Report
Source type
Table
Table
Employee Test [u_employee_test]
Description
There is no description for this table. To add a description, please contact your admin.
Table: Employee Test [u_employee_test]
All
Create your report with Analytics Q&A
Ask for information. You can give simple filtering conditions.
You get the answer with an appropriate visualization.
What do you want to see?
Ask
How can I improve my results?
Next


Create a report
Save
Run
Data
Type
Configure
Style
Report Title :
Employee List Report
list
1 Report types found by search
Select one of the valid types
Other
There are even more chart types to choose from.
List
Type a question about your data
What do you want to see?
Ask
How can I improve my results?
To modify the current report, use the left panel or Edit Condition.
Table: Employee Test [u_employee_test]
All
Employee List Report
Employee ID
Employee Name
Email
Department
Location
SB-0001
Ratan Kumar
test1@gmail.com
ServiceNow
Hyderabad
SB-0002
Sandeep Gujja
test2@gmail.com
ServiceNow
Hyderabad
SB-0003
Meghana Katoju
test3@gmail.com
ServiceNow
Hyderabad
SB-0004
Ajay
test4@gmail.com
Servicenow
Hyderabad
SB-0005
Bitla Sai Prasad
test5@gmail.com
ServiceNow
Hyderabad
SB-0006
Sushma Ganji
test6@gmail.com
ServiceNow
Hyderabad
SB-0007
Yaswanth Nikku
test7@gmail.com
ServiceNow
Hyderabad
1 to 17 of 17
Back
Next





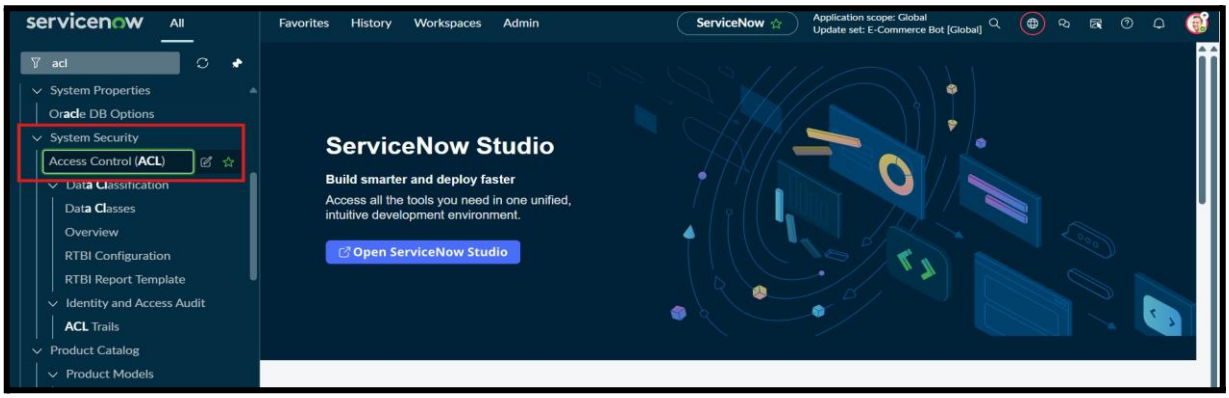
PHASE 9 : Add Reports to Dashboard

CreateDashboard

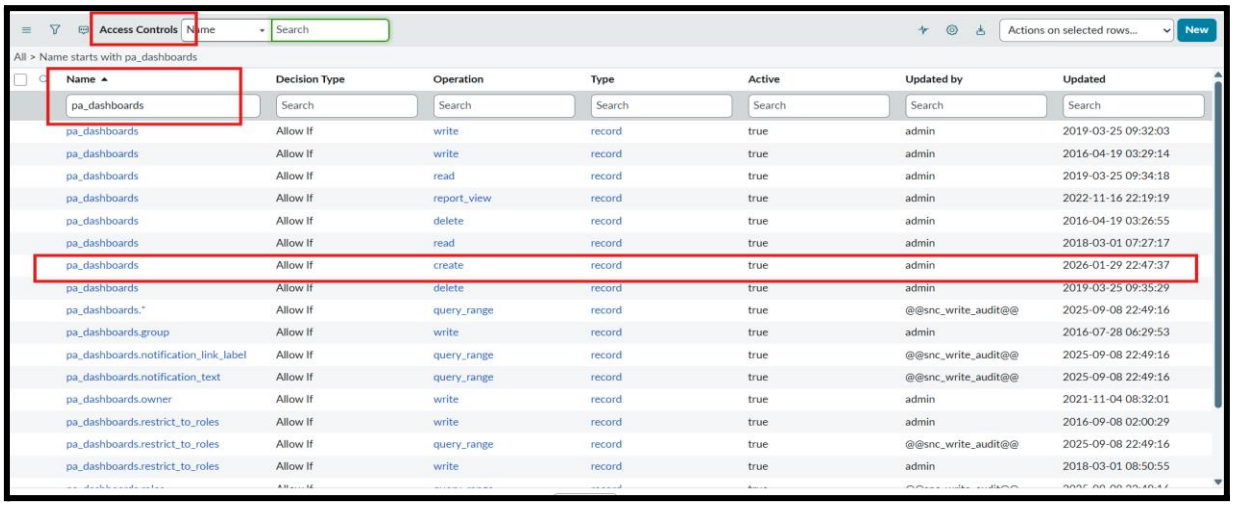
Navigation :

- OpenAllmenus→SearchforDashboards.
- UnderSelfService→ClickDashboards.
- OnthetoprightcornerClickCreateDashboardsor/
- YoucancreateDirectlybyusingthebackendname [PA_DASHBOARDS.FORM].
- ThenClickNewtoCreatenewDashboards.

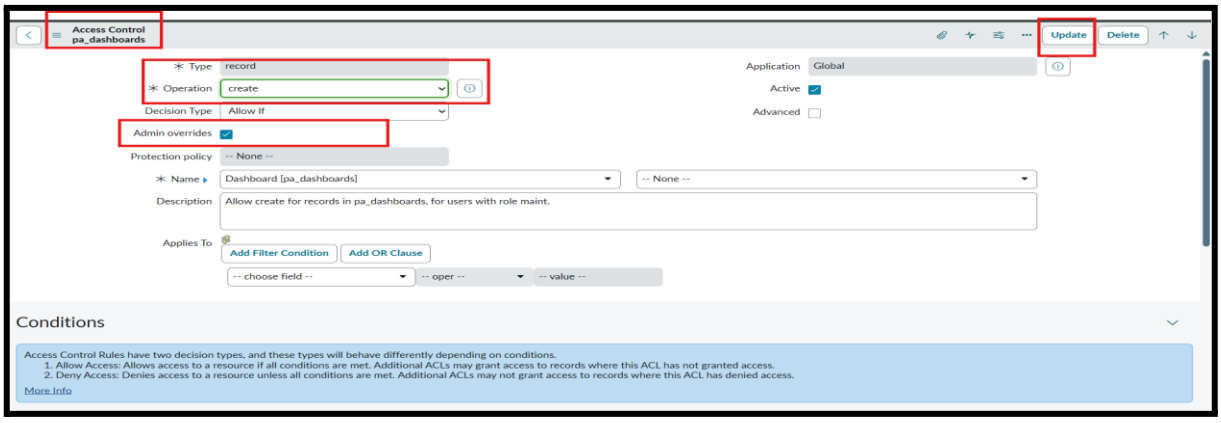
- Onlythebackendnamewillopentheformiftheb**ACL**hasthe**pa_dashboards** access to **Admin Overrides** True only.
- Thenonlywehaveaccesstotheform.



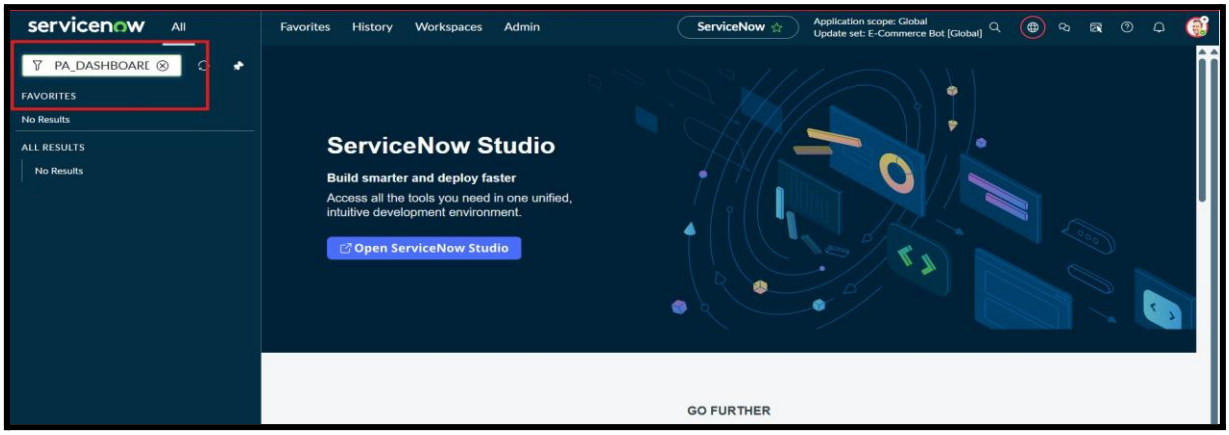
- AfterOpeningtheACLSearchthepa_dashboardsonNameColumn.

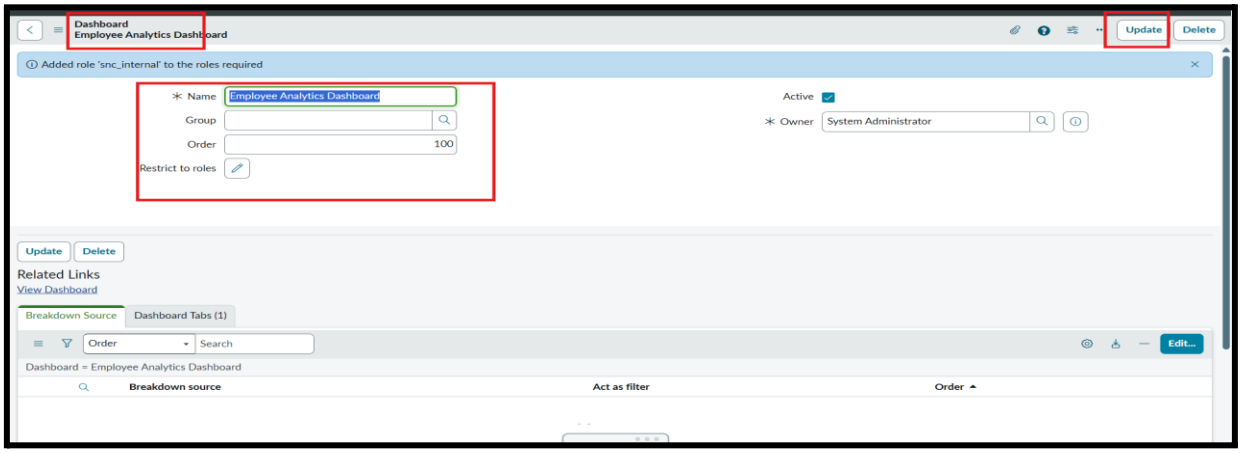


- NowOpenthe**CreateOperation**row.



- ThenClickEnabletheTrueforAdminOverridesandUpdatetheForm.
- Thenthecreationofdashboardswillbepossiblebyusingthebackendname.
- NowOpenthe**PA_DASHBOARDS.FORM**createdashboard.
- Dashboardname:**EmployeeAnalyticsDashboards.**
- **Save**theForm.



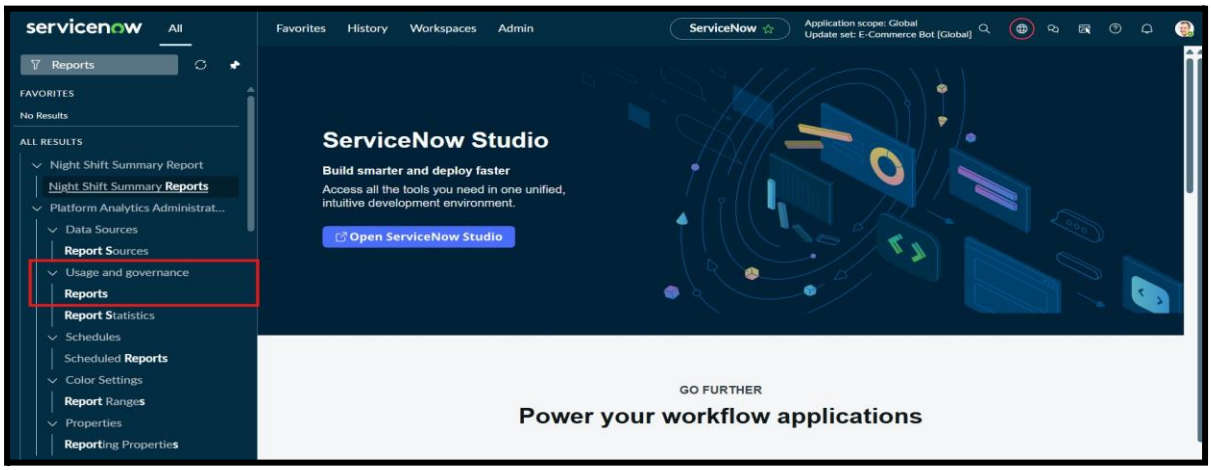


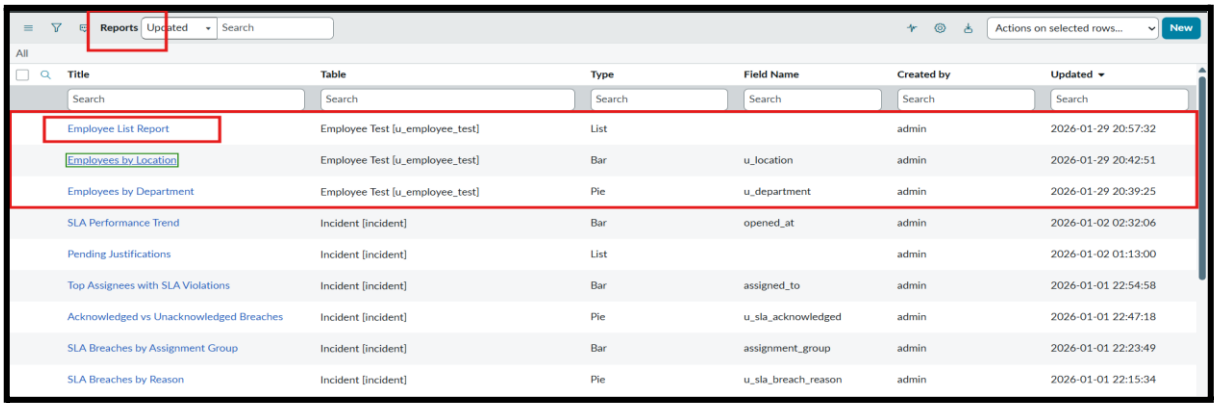
Add Reports to Dashboards.

NavigationSteps:

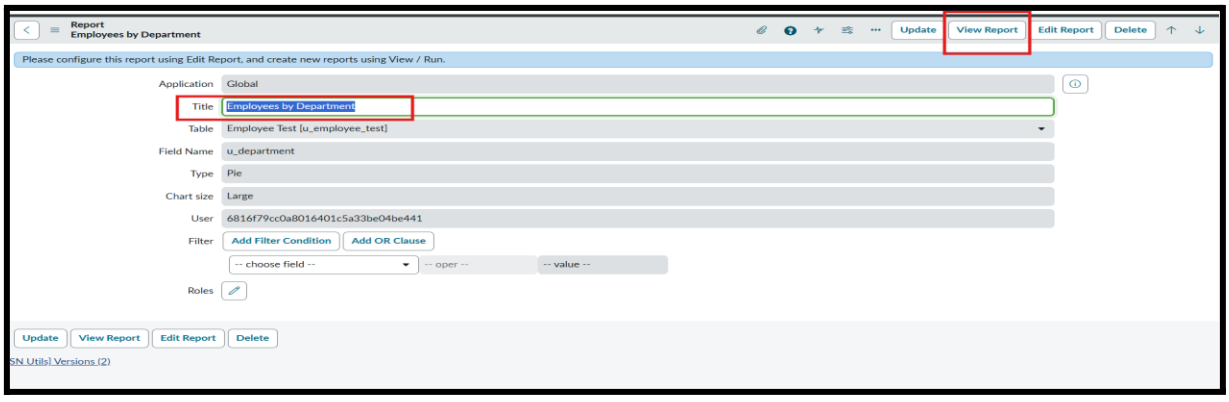
- Openthe**PreviouslyCreatedReports.**
- OnAllmenus→typeReportsUnder**UsageandGovernance**select**reports**.
- Openthe**EmployeesbyDepartment**.
- SelecttheEditReportandClickthe**Share**buttonthentherighttopcornerSelect the **Add to Dashboards** Option.
- ThenChoosetheCreated**dashboardname**andClick**Add**button.

- SameLikethatdotheremaining**2reports**inthesamenavigationhowweadded **Employees by Department.**
- The**EmployeesbyLocation**and**EmployeeListReport**thesetworeportsare also added to the created dashboard.

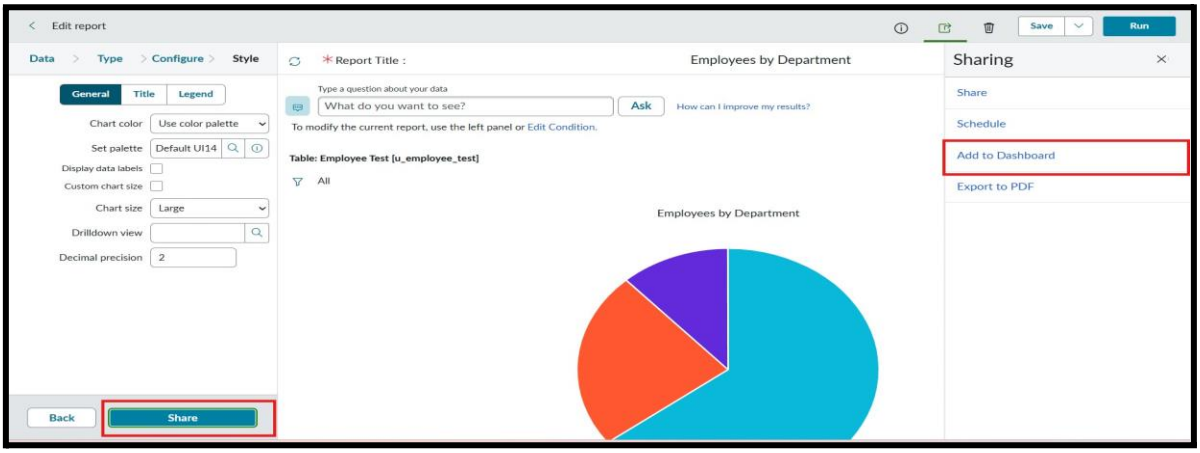


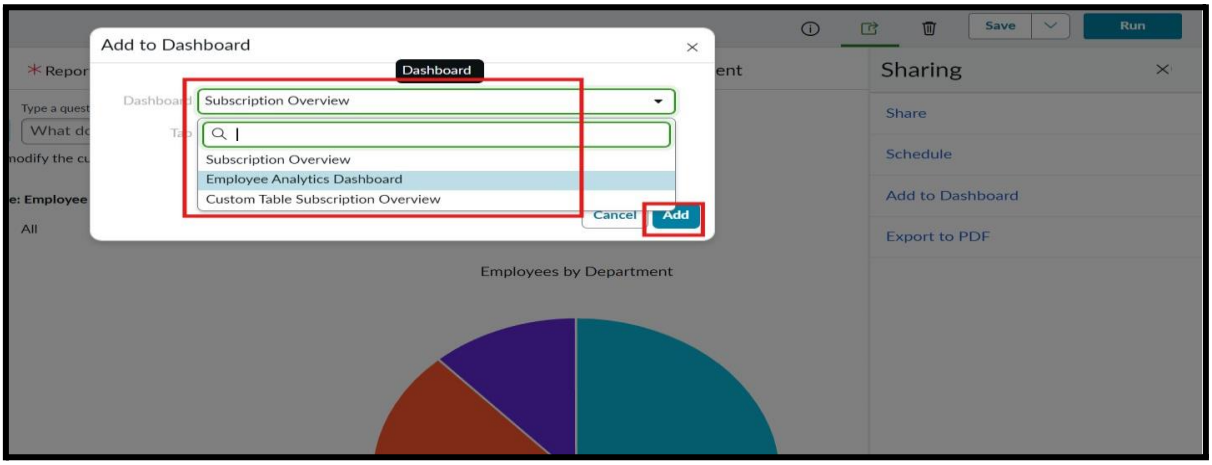


- AfterOpeningtheReportClickthe**ViewReport**Buttonontherightsideofthe Context menu.

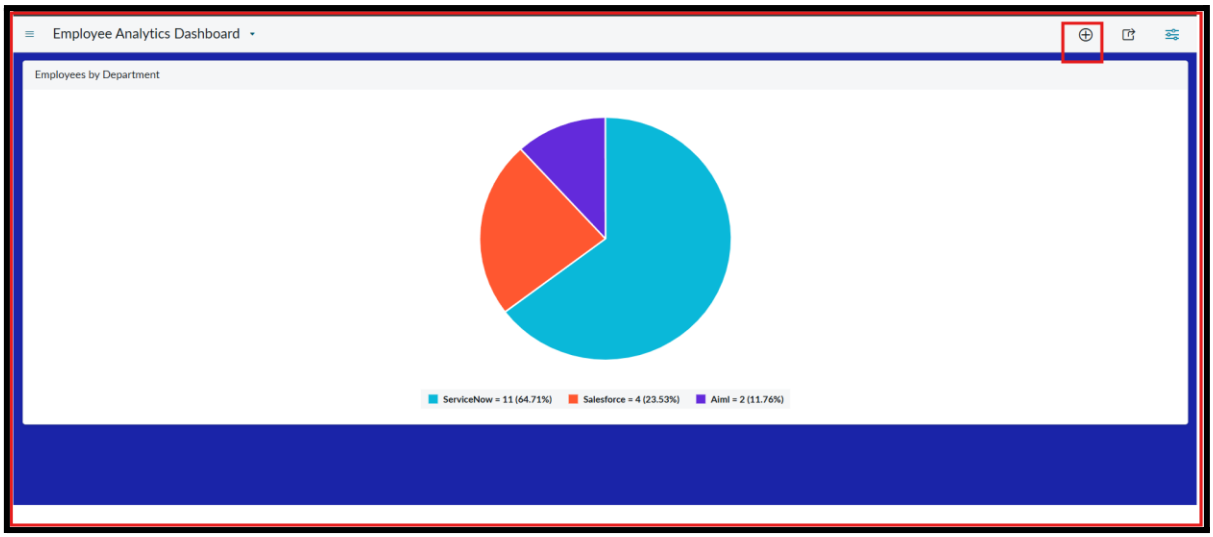


- AfterClickingtheViewReportbutton,NowClickthe**Share**buttonandthen share options are visible on the right side.
- Nowclickthe**AddtoDashboards**.

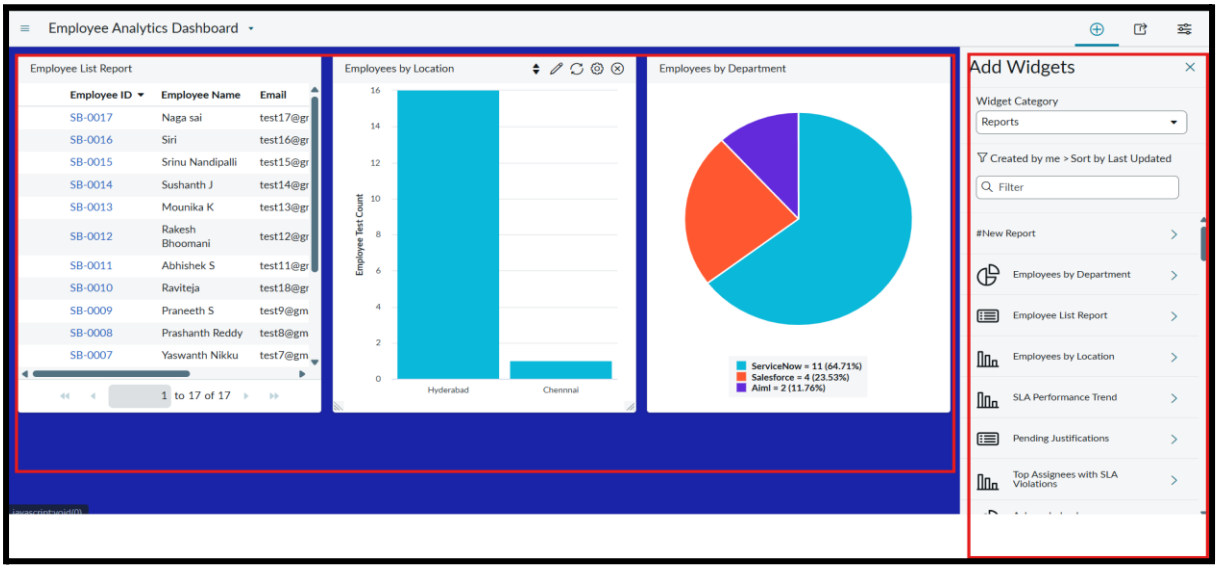




- NowAddtheDashboardthatyoucreatedcalled**EmployeeAnalytics Dashboards.**
- ThenClick**Add**button.
- Nowthe**Report**isaddedtothe**dashboards**.
- Now,Sameintheprocess,addtheremaining**tworeports**tothe**dashboard**.



- Fromthe+iconalsoweaddwidgetsthatourreportsfromthere.
- Ifwewantwecanalsoconfigurethebackgroundcolor.





- HerethethreeReportsareaddedsuccessfullybyvarying**Department,Location and List Report.**
- WecanalsorestricttheUsers,GroupsandRolesFromShareOption.

Conclusion:

1. Thisprojectsuccessfullyimplementedanautomatedemployeedatamanagement solutionusingServiceNowImportSetsandTransformMaps,ensuringaccuracy, efficiency, and data integrity throughout the employee onboarding and update process.
2. AkeyenhancementinthisprojectwastheuseofCoalesceintheTransformMap. ByenablingCoalesceontheEmployeeIDfield,thesystemintelligentlyidentifies existing employee records during every import. This prevents the creation of duplicaterecordsandensuresthatrepeatedimportsupdateexistingdatainsteadof inserting redundant entries. As a result, the employee database remains clean, reliable, and consistent, which is critical in real-world enterprise environments where data may be imported multiple times from external systems.
3. In addition to data integrity, the project was extended with Reports and Dashboardstoprovidemeaningfulinsightsandreal-timevisibilityintoemployee data and import activities. Custom reports were created to track:

- Totalemployeerecords.
- Newlyaddedemployees.
- Updatedemployeerecords.
- Department-wiseemployeedistribution.
- Importsuccessanderrorcounts.

4. These reports were then consolidated into a Dashboard, allowing administrators and HR teams to monitor employee data trends, import performance, and data qualityfromasinglecentralizedview.Dashboardsimproveddecision-makingby presenting information visually through charts and graphs, reducing the need for manual analysis.